BEEE Unit 2 AC circuits Part 1

Course Objectives:

- To provide knowledge of single phase and three phase AC circuits.

Course Outcomes:

- Evaluate and analyze 1-Φ and 3-Φ AC circuits.

CONTENTS

1. Generation of alternating voltage
2. Basic definitions-average and RMS values, phasor and phase difference , sums on phasors
3. Single phase ac series and parallel consisting of – R,L,C,RL,RC, RLC ckts.
4. Definitions-real,reactive and apparent power
5. Admittance Y
6. Series and parallel resonance (only theory)
7. Generation of 3 phase voltages, voltage & current relationships in Star and Delta connections.

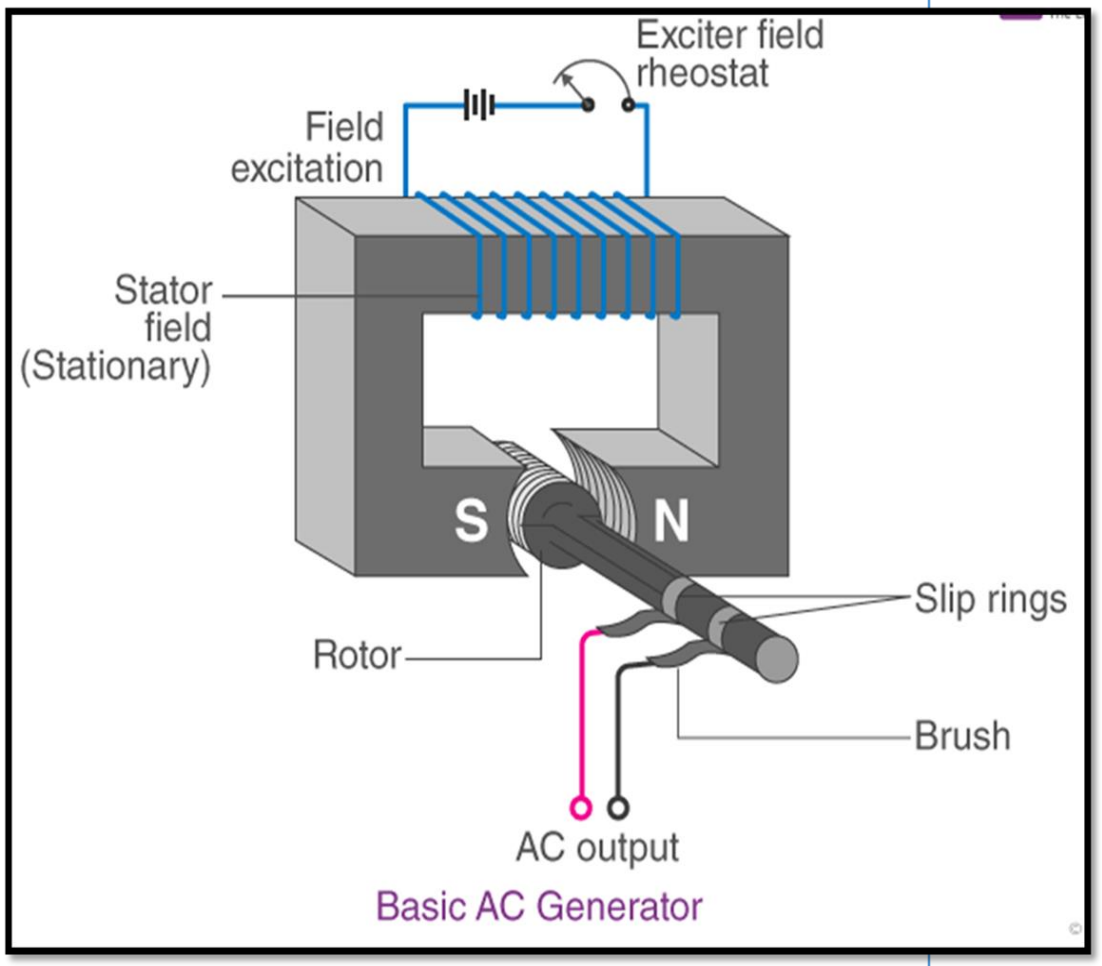
1.

Generation of alternating voltage

AC generator is a machine that converts mechanical energy into electrical energy. The AC Generator's input supply is mechanical energy supplied by steam turbines, gas turbines and combustion engines. The output is an alternating electrical power in the form of alternating voltage and current.

AC generators work on the principle of Faraday's law of electromagnetic induction. When the armature rotates between the magnet's poles upon an axis perpendicular to the magnetic field, the flux linkage of the armature changes continuously. Due to this, an emf is induced in the armature. As a result, an electric current flows through the galvanometer and the slip rings and brushes.

The generated EMF depends on the number of armature coil turns, magnetic field strength, and the rotating field's speed.



- The various parts of an AC generator are:

1.Field- The field consists of coils of conductors that receive a voltage from the source and produce magnetic flux. The magnetic flux in the field cuts the armature to produce a voltage. This voltage is the output voltage of the AC generator.

2. Armature - The part of an AC generator in which the voltage is produced is known as an armature. This component primarily consists of coils of wire that are large enough to carry the full-load current of the generator.

3. Prime Mover -The component used to drive the AC generator is known as a prime mover. The prime mover could either be a diesel engine, a steam turbine, or a motor.

4. Rotor-The rotating component of the generator is known as a rotor. The generator's prime mover drive the rotor.

5. Stator -The stator is the stationary part of an AC generator. The stator core comprises a lamination of steel alloys or magnetic iron to minimise the eddy current losses.

6. Slip Rings -Slip rings are electrical connections used to transfer power to and fro from the rotor of an AC generator. They are typically designed to conduct the flow of current from a stationary device to a rotating one.

Working of an AC Generator -When the armature rotates between the poles of the magnet upon an axis perpendicular to the magnetic field, the flux linkage of the armature changes continuously. As a result, an electric current flows through the galvanometer and the slip rings and brushes. The galvanometer swings between positive and negative values. This indicates that there is an alternating current flowing through the galvanometer. The direction of the induced current can be identified using Fleming's Right-Hand Rule.

Let us consider that, a coil having 'N' no. of turns is rotating in a uniform magnetic field at an angular velocity of ω radian/sec.

It starts from initial position of X-X and rotated bt an angle θ in time 't' second in anticlockwise direction.

$$\theta = \omega t$$

At this instant flux perpendicular to the coil is $\phi_m \cos \omega t$ where ϕ_m is the maximum possible flux.

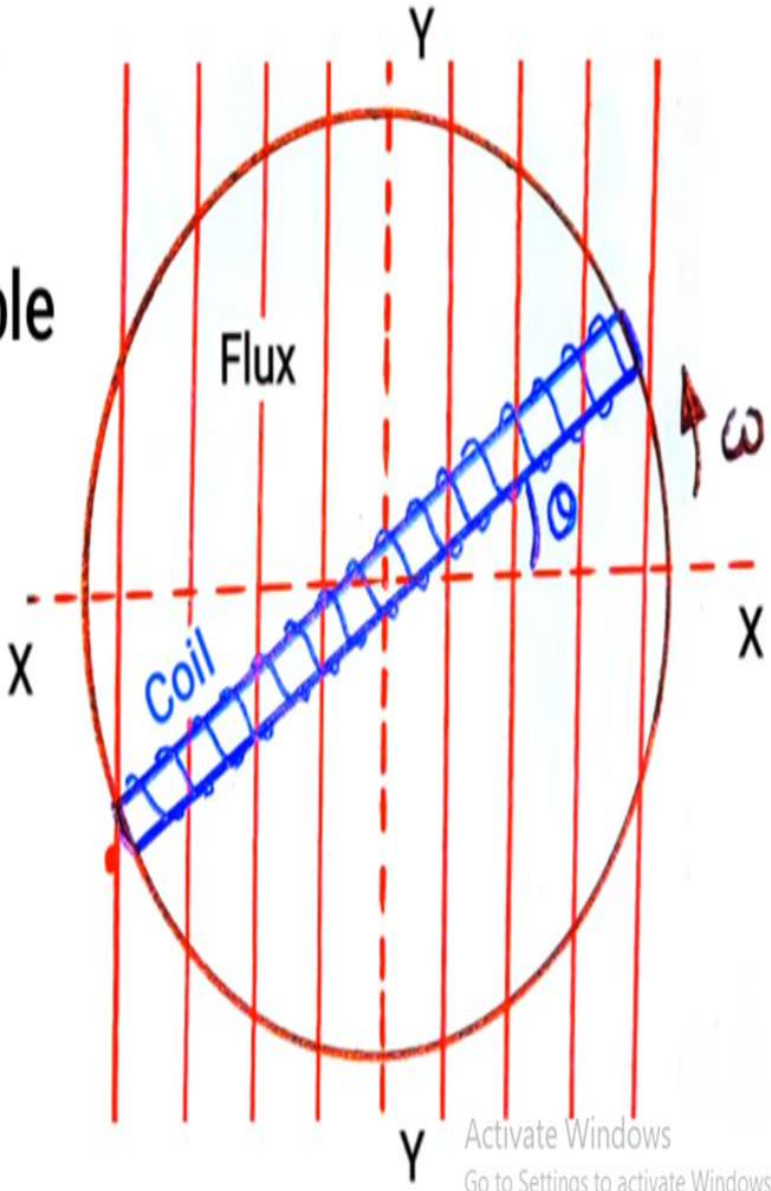
By Faraday's law :

$$E = -N \frac{d\phi}{dt} \qquad E = -N \frac{d}{dt} \phi_m \cos \omega t$$

$$E = N \phi_m \omega \sin \omega t$$

$$E = V_m \sin \omega t$$

$$V_m = N \phi_m \omega = \text{constant}$$

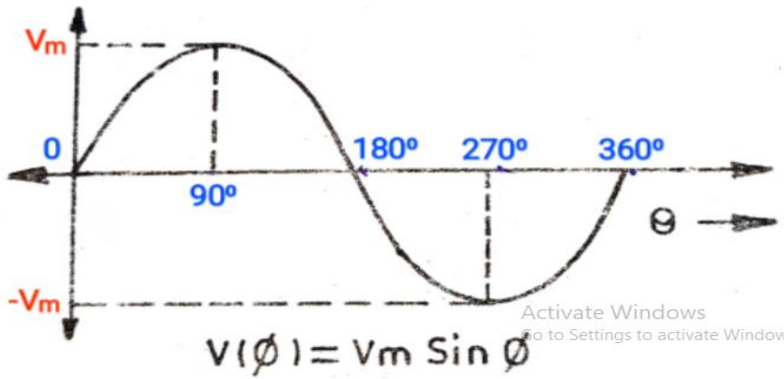


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$$E = V_m \sin \omega t$$

Hence the voltage produce will depend on the value of $\sin \omega t$ and the curve obtained for different values of θ is a "sine-curve".

The EMF induced is therefore known as "sinusoidal EMF".



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Alternating Voltage or EMF,

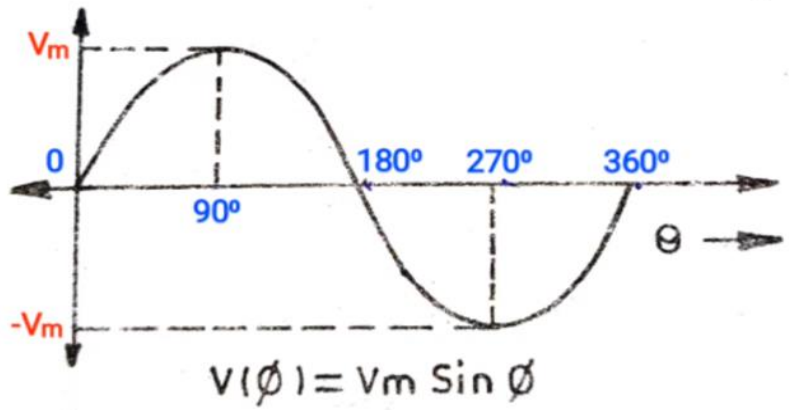
$$E = V_m \sin \omega t$$

$$E = V_m \sin \theta$$

Since $(\theta = \omega t)$

$$E = V_m \sin (2 \pi f t)$$

Since $(\omega = 2 \pi f)$



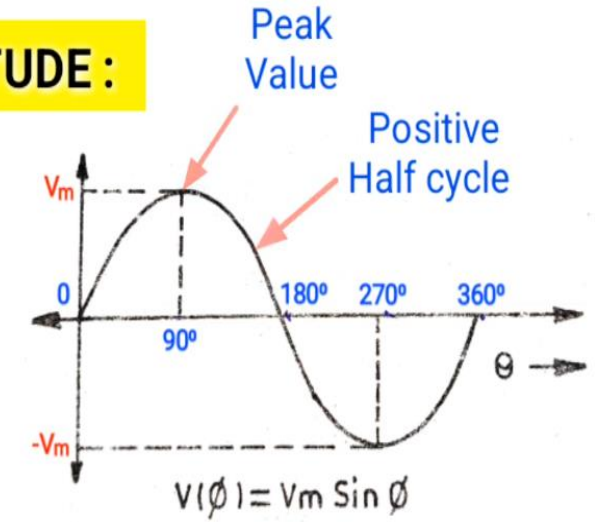
Alternating Current 》》 $I = I_m \sin(\omega t \pm \phi)$

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SOME DEFINITIONS USE IN A.C. FUNDAMENTALS

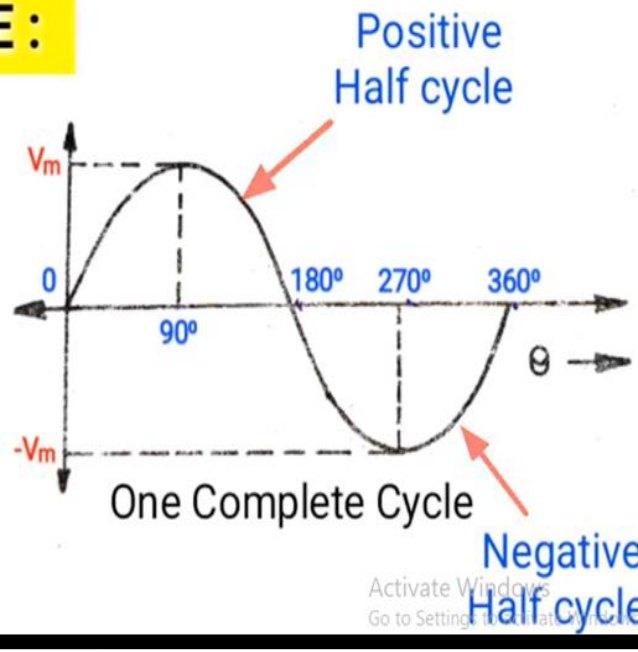
2) AMPLITUDE :

The maximum value (positive or negative) of an alternating quantity in half cycle is known as its amplitude. It is also known as peak value.



3) CYCLE :

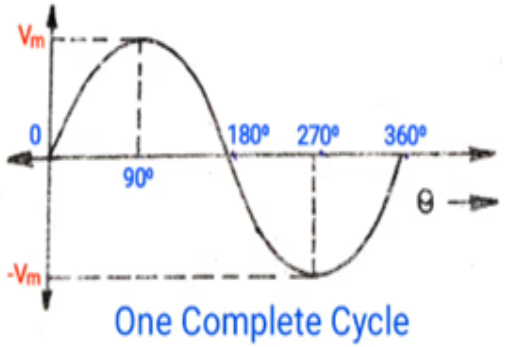
One complete set of positive and negative values of alternating quantity is known as a cycle.



4) TIME PERIOD :

Time required to complete one cycle of an alternating quantity is called time period.

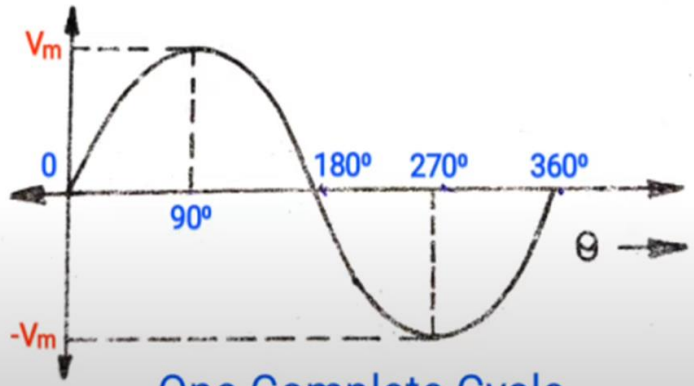
$T = 1 / f$



5) FREQUENCY :

Number of cycles per second is called the frequency of the alternating quantity.

$f = 1 / T$



6) PHASE :

It is part of time-period after the alternating quantity passed through zero position.

OR

It is the phase angle position of A.C. quantity.

7) PHASE DIFFERENCE :

If two alternating quantities do not reach their zero value in the same direction simultaneously then they have phase difference.

The difference of phase angle between two A.C. quantities on the same reference axis is known as phase difference.

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8) ROOT MEAN SQUARE VALUE (RMS VALUE) :

It is that value of D.C. current which when flowing through a given circuit for a given time produces the same heat as produced by the alternating current when flowing through the same circuit for the same time.

$$\text{RMS value} = \frac{\text{Max.value}}{\sqrt{2}}$$

$$\text{RMS value} = 0.707 \times \text{Max.value}$$

$$V_{rms} = \frac{V_m}{\sqrt{2}}$$

$$I_{rms} = \frac{I_m}{\sqrt{2}}$$

9) MEAN VALUE OR AVERAGE VALUE :

The average value of an alternating current is that D.C. current which transfers across any circuit the same charge as it transferred by that alternating current during the same time.

But average value of a symmetrical alternating waveform is always zero because it has equal negative and positive values. Hence average value over a complete cycle is zero. We will calculate average value for half cycle only.

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9) MEAN VALUE OR AVERAGE VALUE :

The average value of an alternating current is that D.C. current which transfers across any circuit the same charge as it transferred by that alternating current during the same time.

$$\text{Avg.value} = \frac{2}{\pi} \times \text{Max.value}$$

Avg.value = 0.637 × Max.value

10) FORM FACTOR :

It is the ratio of RMS value to the Average value.

$$\text{Form Factor} = \frac{\text{RMS value}}{\text{Avg.value}}$$

$$\text{Form Factor} = \frac{\text{Max.value} / \sqrt{2}}{0.637 \times \text{Max.value}}$$

Form Factor = 1.11

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11) CREST FACTOR OR AMPLITUDE FACTOR :

It is the ratio of Maximum value to the RMS value

$$\text{Crest Factor} = \frac{\text{Max.value}}{\text{RMS value}}$$

$$\text{Crest Factor} = \frac{\text{Max.value}}{\frac{\text{Max.value}}{\sqrt{2}}}$$

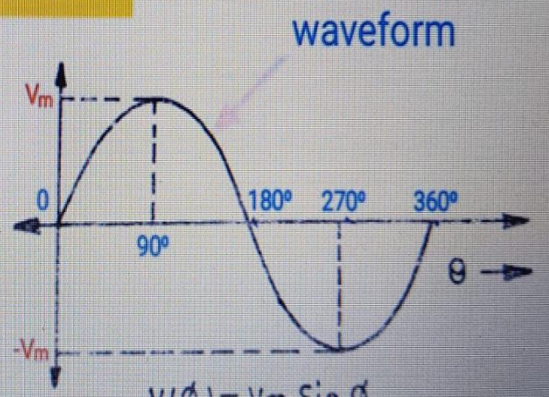
$$\text{C.F.} = \sqrt{2}$$

C.F. = 1.414

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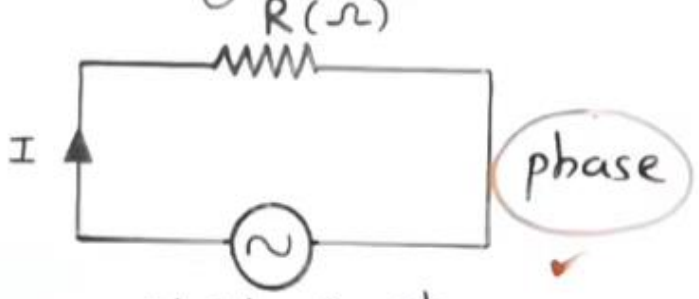
1) WAVEFORM :

The nature of graph of alternating quantity against time is known as waveform or waveshape.



3. Single phase ac series and parallel consisting of – R,L,C,RL,RC, RLC ckts.

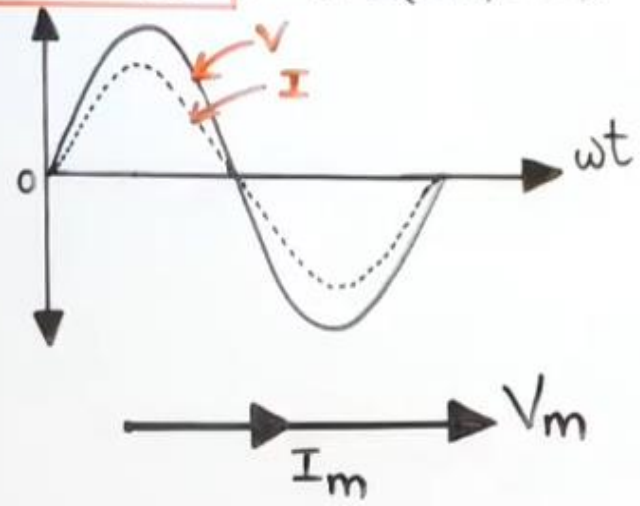
AC through Pure Resistance



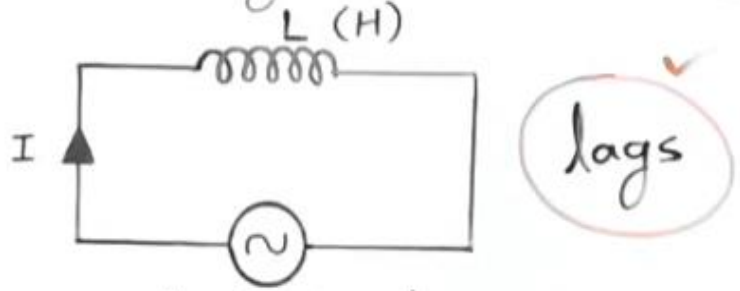
$V = V_m \sin \omega t$

$I = I_m \sin \omega t$

$I_m = \frac{V_m}{R}$, $V = (V_m \angle 0)\ V$, $I = (I_m \angle 0)\ A$



AC through Pure Inductance

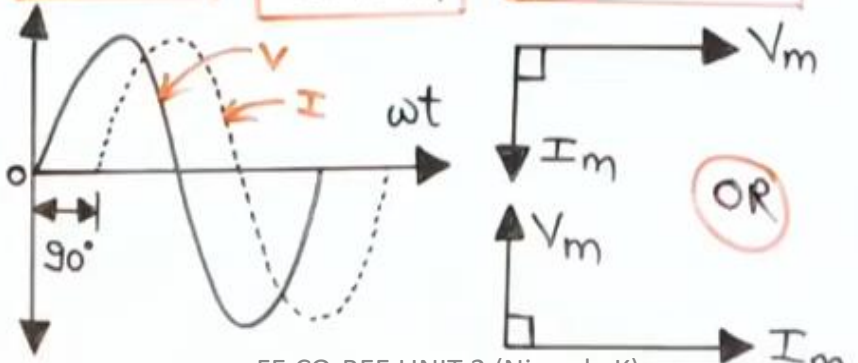


$V = V_m \sin \omega t$

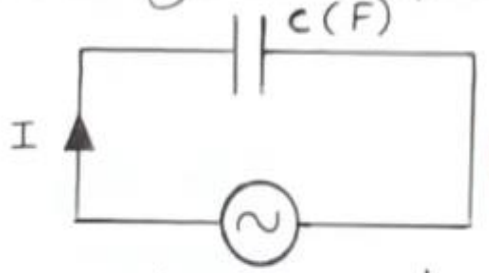
$I = I_m \sin(\omega t - 90^\circ)$

$I_m = \frac{V_m}{X_L}$, $V = (V_m \angle 0)\ V$, $I = (I_m \angle -90^\circ)\ A$

$X_L = \omega L$, $\omega = 2\pi f$, $X_L = 2\pi f L$



AC through Pure Capacitance

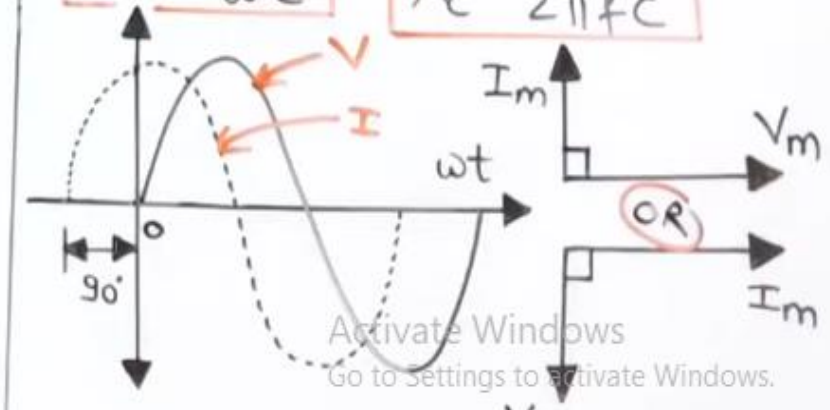


$V = V_m \sin \omega t$

$I = I_m \sin(\omega t + 90^\circ)$

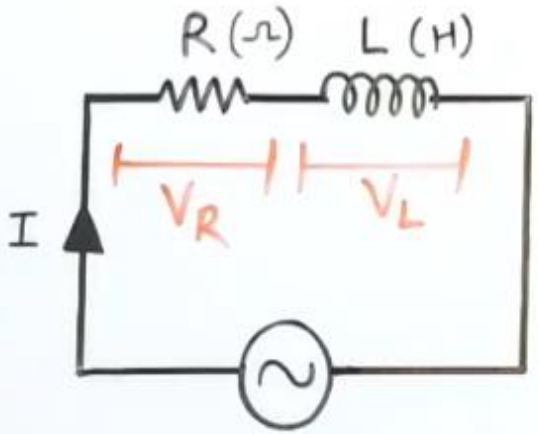
$I_m = \frac{V_m}{X_C}$, $V = (V_m \angle 0)\ V$, $I = (I_m \angle 90^\circ)\ A$

$X_C = \frac{1}{\omega C}$, $X_C = \frac{1}{2\pi f C}$



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Series AC Circuit : Series R-L Circuit



$$I = \frac{V}{Z} \text{ Amp.}$$

where,

Z = Impedance (Ω)

$$Z = R + jX_L$$

OR

$$Z = \sqrt{R^2 + X_L^2}$$

$$V = V_m \sin \omega t$$

$$I = I_m \sin(\omega t - \phi)$$

$$V = (V_m \angle 0)\ V$$

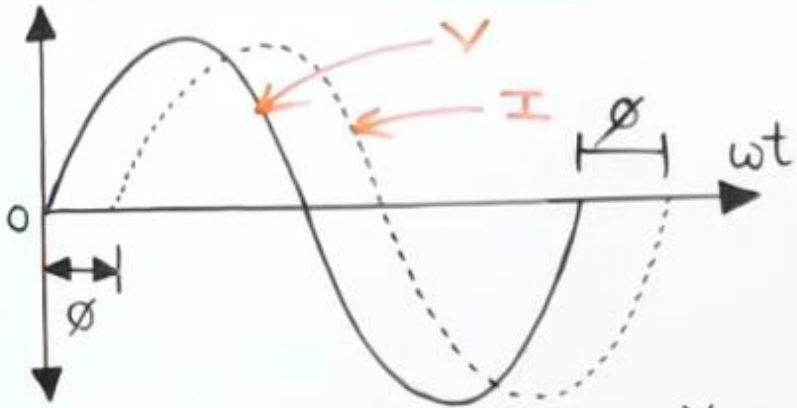
$$I = (I_m \angle -\phi)\ A$$

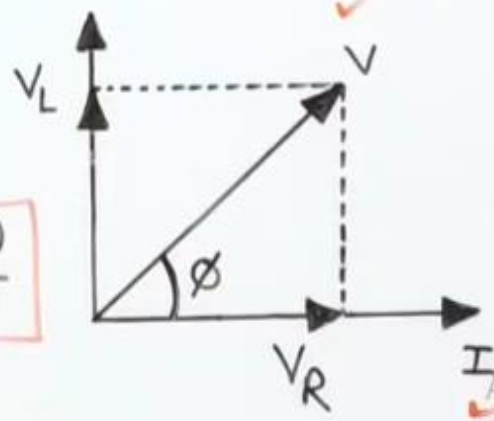
$$X_L = \omega L$$

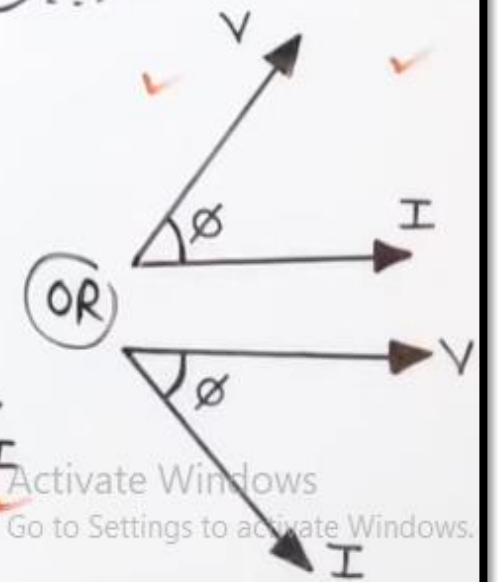
$$X_L = 2\pi f L$$

$$\because \omega = 2\pi f$$

$$\phi = \tan^{-1}\left(\frac{X_L}{R}\right)$$

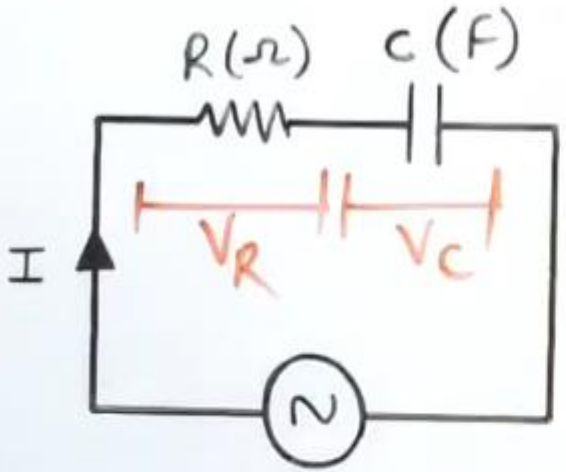






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Series AC circuits : Series R-C circuit



$$I = \frac{V}{Z} \quad \text{Amp}$$

where,

Z = Impedance (Ω)

$$Z = R - jX_C$$

OR

$$Z = \sqrt{R^2 + X_C^2}$$

$$V = V_m \sin \omega t$$

$$I = I_m \sin(\omega t + \phi)$$

$$V = (V_m \angle 0)\ V$$

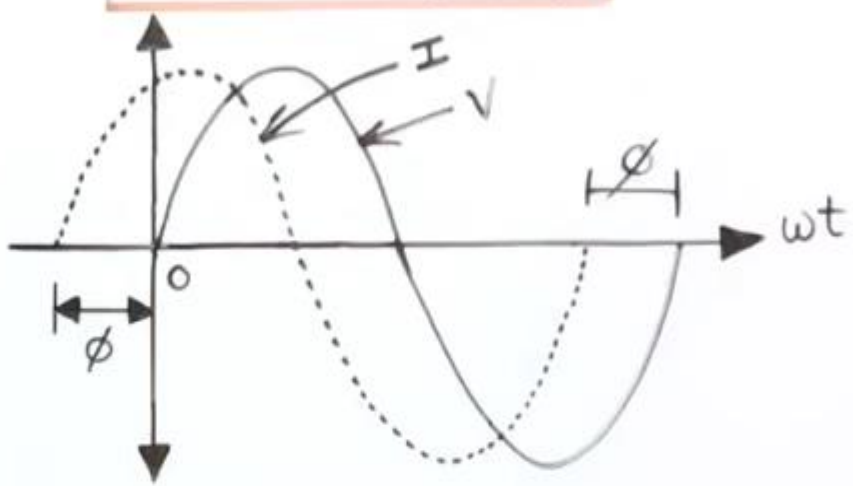
$$I = (I_m \angle \phi)\ A$$

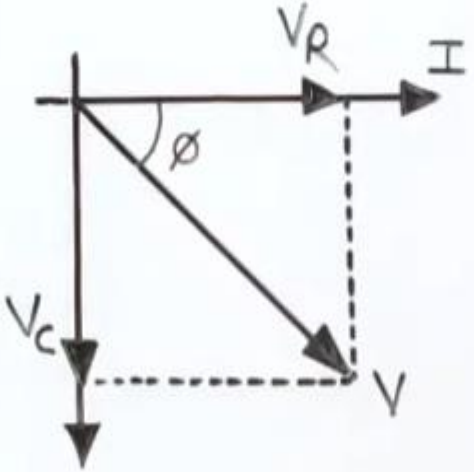
$$X_C = \frac{1}{\omega C}$$

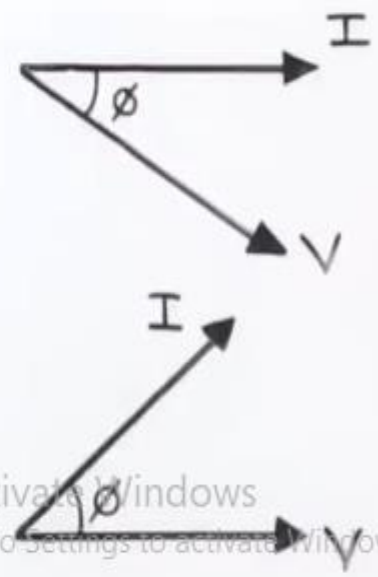
$$X_C = \frac{1}{2\pi f C}$$

$\because \omega = 2\pi f$

$$\phi = \tan^{-1}\left(\frac{-X_C}{R}\right)$$

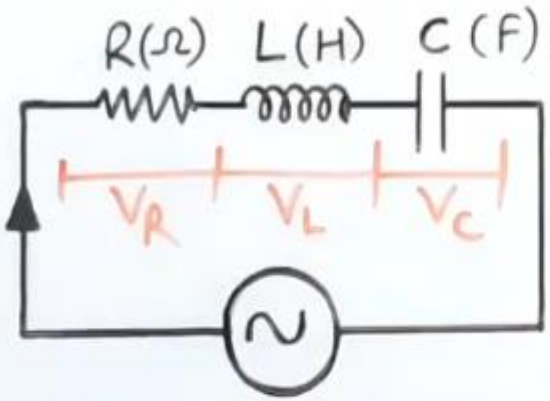






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- Series AC circuit : Series RLC circuit -



$$Z = R + j(X_L - X_C) \quad (\Omega)$$

OR

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$\phi = \tan^{-1}\left(\frac{X_L - X_C}{R}\right)$$

$$V = V_m \sin \omega t$$
$$I = I_m \sin(\omega t \pm \phi)$$

$$V = (V_m \angle 0)\ V$$
$$I = (I_m \angle \pm \phi)\ A$$

$$I = \frac{V}{Z} \text{ Amp.}$$

where, Z = Impedance (Ω)

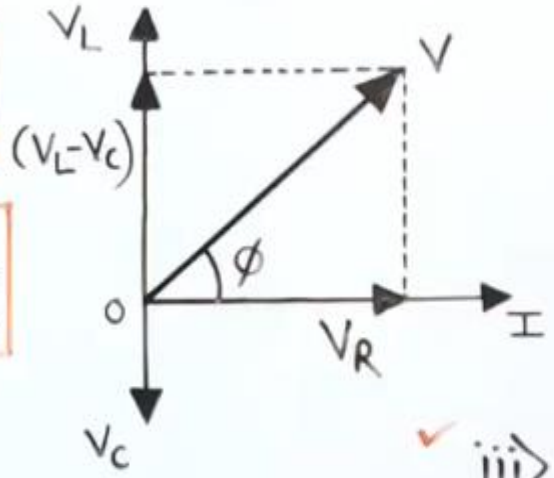
$$X_L = \omega L \qquad X_C = \frac{1}{\omega C}$$

$\because \omega = 2\pi f$

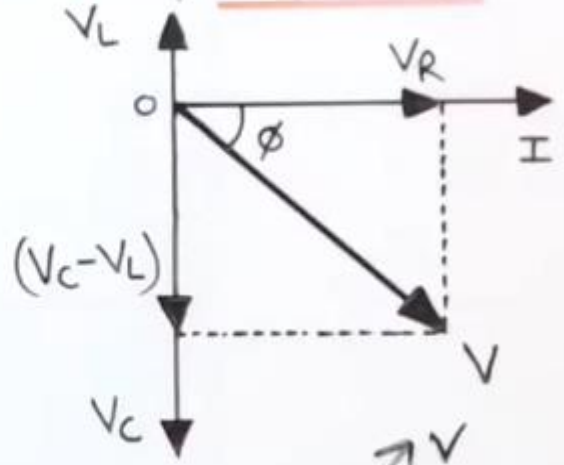
$$X_L = 2\pi f L \quad (\Omega)$$

$$X_C = \frac{1}{2\pi f C} \quad (\Omega)$$

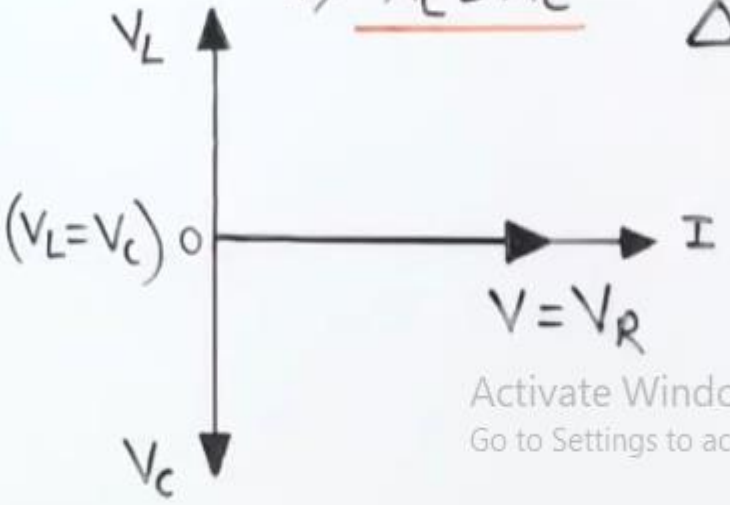
i) $X_L > X_C$

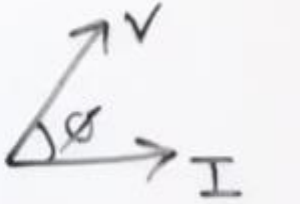


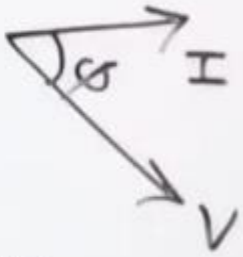
ii) $X_C > X_L$



iii) $X_L = X_C$







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1-phase AC circuit formulae

$V = V_m \sin \omega t$ V

$I = I_m \sin(\omega t \pm \phi)$ A

For R ckt: $I = V/R$ A

For L ckt: $I = V/X_L$ A

For C ckt: $I = V/X_C$ A

$X_L = \omega L$, $X_C = \frac{1}{\omega C}$ Ω

$X_L = 2\pi f L$, $X_C = \frac{1}{2\pi f C}$ Ω

$\omega = 2\pi f$

* For R-L ckt: $I = V/Z$ A

$Z = R + jX_L$, $Z = \sqrt{R^2 + X_L^2}$ Ω

$\phi = \tan^{-1}\left(\frac{X_L}{R}\right)$

* For R-C ckt: $I = V/Z$ A

$Z = R - jX_C$, $Z = \sqrt{R^2 + X_C^2}$

$\phi = \tan^{-1}\left(\frac{-X_C}{R}\right)$

* For R-L-C ckt: $I = V/Z$ A

$Z = R + j(X_L - X_C)$ Ω

$Z = \sqrt{R^2 + (X_L - X_C)^2}$ Ω

$\phi = \tan^{-1}\left(\frac{X_L - X_C}{R}\right)$

* Power factor = $\cos\phi$

(lagging or leading)

$V_R = I \times R$

$V_L = I \times (X_L \angle 90)$ V

$V_C = I \times (X_C \angle -90)$ V

* Active Power, $P = I^2 R$ W or KW

$P = |V| \cdot |I| \cdot \cos\phi$ W or KW

* Reactive Power, $Q = |V| \cdot |I| \cdot \sin\phi$ VAR KVAR

* Apparant Power, $S = \sqrt{P^2 + Q^2}$ VA or KVA

$V_{rms} = V_m/\sqrt{2}$ $I_{rms} = I_m/\sqrt{2}$

Z = Impedance X = Reactance

X_L = Inductive Reactance

X_C = Capacitive Reactance

ω = angular velocity

t = time f = frequency

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Given:

$V = 325 \sin \omega t$

$I = 7 \sin (\omega t + 63^\circ)$

$V_{rms} = ?$

$I_{rms} = ?$

$Z = ?$

$\phi = ?$

The sinusoidal voltage applied to a circuit of unknown impedance is V = 325 sin wt and the current I = 7 sin (wt + 63°). Hence find 1) RMS value of voltage and current.
2) Unknown impedance.
3) Power factor angle.
Comment whether the current is lagging or leading w.r.t. voltage.

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Given:

- $V = 325 \sin \omega t$
- $I = 7 \sin(\omega t + 63^\circ)$
- $V_{rms} = ?$
- $I_{rms} = ?$
- $Z = ?$
- $\phi = ?$

$I = \frac{V}{Z}$ $\quad Z = \frac{V}{I}$

$\Rightarrow$ we have,

$V = 325 \sin \omega t$

compair it with $V = V_m \sin \omega t$

$\therefore V_m = 325 \text{ V}$

also we have,

$I = 7 \sin(\omega t + 63^\circ)$

Compair it with $I = I_m \sin(\omega t \pm \phi)$

$\therefore I_m = 7 \text{ A}$, $\phi = 63^\circ$

$\therefore V_{rms} = V_m / \sqrt{2}$

$= 325/\sqrt{2}$

$V_{rms} = 229.809 \text{ V}$

and $I_{rms} = I_m / \sqrt{2}$

$= 7/\sqrt{2}$

$I_{rms} = 4.949 \text{ A}$

Impedance, $Z = \frac{V}{I}$

$Z = \frac{V_{rms}}{I_{rms}}$, $Z = \frac{V_m}{I_m}$

$= \frac{229.809}{4.949}$, $= \frac{325}{7}$

$Z = 46.435 \Omega$ $\quad Z = 46.428 \Omega$

Power factor angle $= \phi = 63^\circ$

* Since the given phase angle 63° is +ve, therefore current is leading the voltage by 63°.

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A series R-L circuit having R=20 ohm and L=0.05 H is connected across 100 V, 50 Hz supply. Calculate circuit current and the power consumed.

Given

R-L ckt

$R = 20\,\Omega$

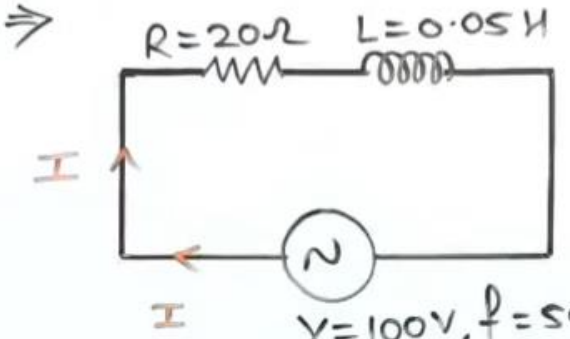
$L = 0.05\ H$

$V = 100\ V$

$f = 50\ Hz$

$I = ?$

$P = ?$



$X_L = 2\pi f L = 2\pi \times 50 \times 0.05$

$X_L = 15.707\,\Omega$

For R-L ckt, $Z = R + jX_L$

$\therefore Z = (20 + j\,15.707)\,\Omega$

$Z = (25.43 \angle 38.144)\,\Omega$

$\therefore$ Current, $I = \frac{V}{Z}$

$= \frac{100}{25.43 \angle 38.144}$

$I = 3.932 \angle -38.144\ A$

Power, $P = |V||I| \cos\phi$

$P = 100 \times 3.932 \cos 38.144$

$P = 309.236\ W$

OR $P = I^2 R$

$P = 3.932^2 \times 20$

$P = 309.212\ W$

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A series R-C circuit has R = 20 ohm and C = 125 microFarad. A voltage of 200 V, 50 Hz is applied to the circuit. Calculate the current and the power lost.

① $X_C = \frac{1}{2\pi f C}$

③ $I = \frac{V}{Z}$

② $Z = R - jX_C$

$Z = \sqrt{R^2 + X_C^2}$, $\phi = \tan^{-1}\left(\frac{-X_C}{R}\right)$

④ $P = |V|\,|I| \cos\phi$

$P = I^2 R$

Given:

R-C ckt

$R = 20\,\Omega$

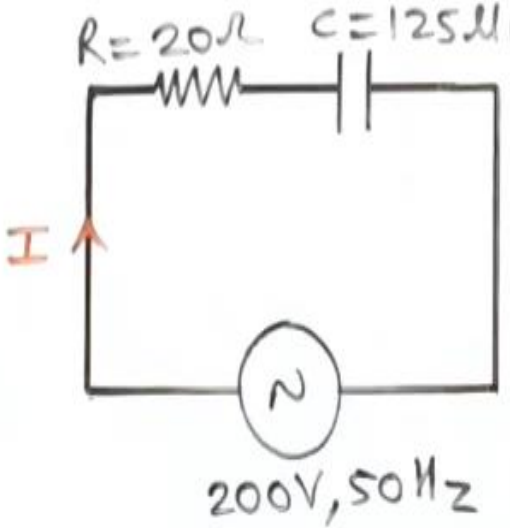
$C = 125\,\mu F = 125 \times 10^{-6}\,F$

$V = 200\,V$

$f = 50\,Hz$

$I = ?$

$P = ?$



$X_C = \frac{1}{2\pi f C}$

$= \frac{1}{2\pi \times 50 \times 125 \times 10^{-6}}$

$X_C = 25.46\,\Omega$

for R-C ckt, $Z = R - jX_C$

$Z = (20 - j25.46)\,\Omega$

$Z = (32.37 \angle -51.848)\,\Omega$

Current, $I = \frac{V}{Z} = \frac{200}{20 - j25.46}$

$I = 6.18 \angle 51.848\,A$

Power, $P = |V|\,|I| \cos\phi$

$= 200 \times 6.18 \cos 51.848$

$P = 763.538\,W$

OR

$P = I^2 R$

$= 6.18^2 \times 20$

$P = 763.848\,W$

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A series R-L-C circuit having R=10 ohm, L=0.2 H and C=100 microF is connected across 100V, 50Hz supply Calculate current and power factor.

Given:

R-L-C ckt.

$R = 10\ \Omega$

$L = 0.2\ H$

$C = 100\ \mu F = 100 \times 10^{-6}\ F$

$V = 100\ V$

$f = 50\ Hz$

$I = ?$

$\cos\phi = ?$

$\Rightarrow X_L = 2\pi f L = 2\pi \times 50 \times 0.2$

$X_L = 62.83\ \Omega$

$X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi \times 50 \times 100 \times 10^{-6}}$

$X_C = 31.83\ \Omega$

For R-L-C ckt, $Z = R + j(X_L - X_C)$

$\therefore Z = 10 + j(62.83 - 31.83)$

$Z = (10 + j31)\ \Omega$

$Z = (32.57 \angle 72.12)\ \Omega$

Current, $I = \frac{V}{Z}$

$I = \frac{100}{32.57 \angle 72.12}$

$I = (3.07 \angle -72.12)\ A$

Power factor $= \cos\phi$

$= \cos 72.12$

P.f. $= 0.307$ (lagging)

$Z = \sqrt{R^2 + (X_L - X_C)^2}$

$\phi = \tan^{-1}\left(\frac{X_L - X_C}{R}\right)$

(4) $I = \frac{V}{Z}$

(3) $Z = R + j(X_L - X_C)$

(1) $X_L = 2\pi f L$ (2) $X_C = \frac{1}{2\pi f C}$

A choke coil is connected across 200V, 50Hz supply. Power consumed by the coil is 960 W and current is 8 A. Calculate circuit constants.

$Z = R + jX_L$

3. $Z = \sqrt{R^2 + X_L^2}$

1. $P = I^2R$

4. $X_L = 2\pi fL$

2. $I = \frac{V}{Z}$

Given:

coil (R-L ckt)

$V = 200V$

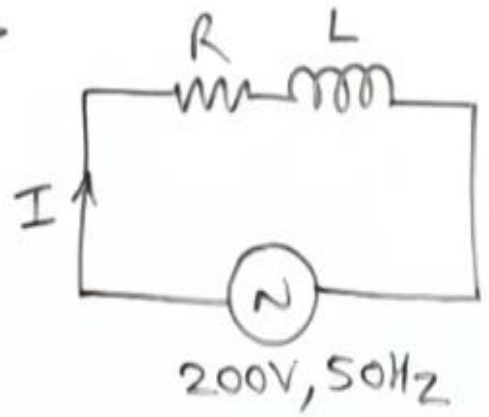
$f = 50Hz$

$P = 960W$

$I = 8A$

R = ?

L = ?



Power,

$P = I^2R$

$960 = 8^2 \times R$

$\therefore R = \frac{960}{64} = 15\Omega$

$\therefore R = 15\Omega$

Current, $I = \frac{V}{Z}$

$\therefore Z = \frac{V}{I} = \frac{200}{8} = 25\Omega$

$\therefore Z = 25\Omega$

For R-L ckt

$Z = \sqrt{R^2 + X_L^2}$

$Z^2 = R^2 + X_L^2$

$\therefore X_L = \sqrt{Z^2 - R^2}$

$= \sqrt{25^2 - 15^2}$

$\therefore X_L = 20\Omega$

$\therefore 2\pi fL = 20\Omega$

$2\pi \times 50 \times L = 20$

$\therefore L = \frac{20}{2\pi \times 50}$

$\therefore L = 0.0636H$

$Z = R + jX_L$

3. $Z = \sqrt{R^2 + X_L^2}$

1. $P = I^2R$

4. $X_L = 2\pi fL$

2. $I = \frac{V}{Z}$

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Series R-C circuit is connected to 200V, 50Hz supply. Circuit consumes 2048Watt and takes a current of 16Amp. Calculate circuit constants.

Given

R-C ckt.

$V = 200V$

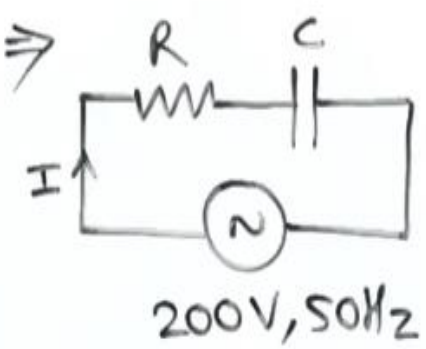
$f = 50Hz$

$P = 2048\,W$

$I = 16\,A$

R = ?

C = ?



Power,

$P = I^2R$

$2048 = 16^2 \times R$

$\therefore R = \frac{2048}{16^2}$

$R = 8\Omega$

Current, $I = \frac{V}{Z}$

$\therefore Z = \frac{V}{I} = \frac{200}{16} = 12.5\Omega$

$\therefore Z = 12.5\Omega$

For R-C ckt

$Z = \sqrt{R^2 + X_C^2}$

$Z^2 = R^2 + X_C^2$

$\therefore X_C = \sqrt{Z^2 - R^2}$

$= \sqrt{12.5^2 - 8^2}$

$X_C = 9.6\Omega$

we have, $X_C = \frac{1}{2\pi fC}$

$\therefore C = \frac{1}{2\pi f X_C} = \frac{1}{2\pi \times 50 \times 9.6}$

$C = 3.315 \times 10^{-4}\,F$

$C = 331.5 \times 10^{-6}\,F$

$C = 331.5\,\mu F$

$Z = R - jX_C$

3. $Z = \sqrt{R^2 + X_C^2}$

1. $P = I^2R$

4. $X_C = \frac{1}{2\pi fC}$

2. $I = \frac{V}{Z}$

$Z = R - jX_C$

3. $Z = \sqrt{R^2 + X_C^2}$

1. $P = I^2R$

4. $X_C = \frac{1}{2\pi fC}$

2. $I = \frac{V}{Z}$

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A choke coil having a resistance of 6 ohm and an inductance of 0.2 H is connected in series with capacitance of 30 microF and voltage of V=141.4sin157t is applied to this RLC combination. Find the energy consume in 3 hrs.

Given:

RLC ckt.

$R = 6\Omega$

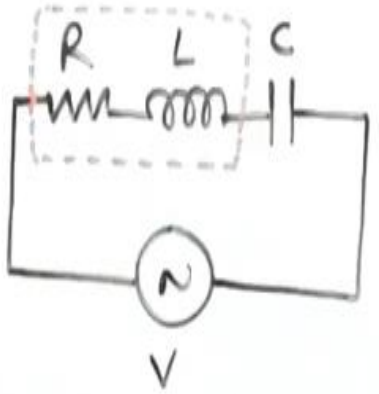
$L = 0.2H$

$C = 30 \times 10^{-6} F$

$V = 141.4 \sin 157t$

$t = 3$ hrs.

Energy = ?



Given that,

$V = 141.4 \sin 157t$

compair it with

$v = V_m \sin \omega t$

$\therefore V_m = 141.4V$, $\omega = 157$

$V_{rms} = \frac{V_m}{\sqrt{2}} = \frac{141.4}{\sqrt{2}}$

$\therefore V_{rms} = 99.984V$

$\omega = 2\pi f$, $f = \omega/2\pi = 157/2\pi = 24.987$

$X_L = 2\pi f L = 2\pi \times 24.987 \times 0.2 = 31.4\Omega$

$X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi \times 24.987 \times 30 \times 10^{-6}} = 212.316$

For RLC ckt, $Z = R + j(X_L - X_C)$

$Z = 6 + j(31.4 - 212.316)$

$Z = (6 - j180.916)\Omega$

Current, $I = \frac{V}{Z} = \frac{99.984}{6 - j180.916}$

$I = 0.552 \angle 88.1$ A

Power, $P = |V|\,|I| \cos\phi$

$= 99.984 \times 0.552 \cos 88.1$

$P = 1.829$ W

Energy $= P \times t$

$= 1.829 \times 3$ hrs

Energy $= 5.487$ W-hr

① $X_L = 2\pi f L$ ③ $X_C = \frac{1}{2\pi f C}$

⑥ Energy $= P \times t$

⑤ $P = |V|\,|I| \cos\phi$

④ $I = \frac{V}{Z}$

③ $Z = R + j(X_L - X_C)$

$Z = \sqrt{R^2 + (X_L - X_C)^2}$

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The voltage and current equation in an A.C. circuit are given by:

- $V = 150 \sin \omega t$
- $I = 3 \sin(\omega t + \pi/2)$

Find the R.M.S. value of voltage and current. Also state the type of circuit and the power factor of the circuit.

Given:

$V = 150 \sin \omega t$

$I = 3 \sin(\omega t + 90)$

To find:

$V_{rms} = ?$ $I_{rms} = ?$

Type of ckt = ?

Power factor = ?

Sol: Given eqn of voltage

$V = 150 \sin \omega t$

compair it with std. eqn

$V = V_m \sin \omega t$

$\therefore V_m = 150 \text{ V}$

$$V_{rms} = \frac{V_m}{\sqrt{2}} = \frac{150}{\sqrt{2}}$$

$V_{rms} = 106.066 \text{ V}$

Also given that,

$I = 3 \sin(\omega t + \pi/2)$

compair it with std. eqn

$I = I_m \sin(\omega t \pm \phi)$

$\therefore I_m = 3 \text{ A}$, $\phi = \pi/2$

$\phi = 90^\circ$

$$I_{rms} = \frac{I_m}{\sqrt{2}} = \frac{3}{\sqrt{2}}$$

$I_{rms} = 2.121 \text{ A}$

Since, $\phi = +90^\circ$

Therefore we can say that, current leads the applied voltage by 90°.

And the given circuit is purely capacitive circuit.

power factor $= \cos \phi = \cos 90^\circ = 0$

It is a leading power factor

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If $v = 100\sin 314t$ ac supply is available, then find:

i) Maximum Value

ii) Angular frequency

iii) Frequency

iv) Time period

v) RMS value

vi) Average Value.

Given:

$v = 100\sin 314t$

To find:

i) $V_m = ?$

ii) $\omega = ?$

iii) $f = ?$

iv) $T = ?$

v) $V_{rms} = ?$

vi) $V_{avg} = ?$

⇒ Given that:

$v = 100\sin 314t$

compair it with std. form

$\therefore v = V_m \sin \omega t$

i) maximum Voltage,

$V_m = 100\ V$

ii) Angular frequency,

$\omega = 314\ rad/sec$

iii) Frequency $\omega = 2\pi f$

$\therefore f = \frac{\omega}{2\pi} = \frac{314}{2\pi}$

$f = 49.974\ Hz$

$f = 50\ Hz$

iv) Time period, $T = \frac{1}{f}$

$= \frac{1}{50}$

$T = 0.02\ sec$

v) RMS value of voltage:

$V_{rms} = \frac{V_m}{\sqrt{2}} = \frac{100}{\sqrt{2}}$

$V_{rms} = 70.71\ V$

vi) Average Value: $V_{avg} = 0.637 \times V_m$

$= 0.637 \times 100$

$V_{avg} = 63.7\ V$

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If $v = 141.4\sin(628t - \pi/6)$ V

then calculate:

i) R.M.S. value

ii) Maximum value

iii) Angular frequency

iv) Frequency

v) Form factor

vi) V at $t = 0.015$ sec.

⇒ Given:

$v = 141.4\sin(628t - \pi/6)$

✓ $V_{rms} = ?$ $V_m = ?$ ✓

✓ $\omega = ?$ $f = ?$ ✓

✓ form factor $= ?$ ✓

✓ $v = ?$ $t = 0.015$ sec.

⇒ We have:

$v = 141.4\sin(628t - \pi/6)$

Compair it with

$v = V_m \sin(\omega t - \phi)$

$\therefore$ $V_m = 141.4$ V ✓

$\omega = 628$ rad/sec. ✓

$\phi = \pi/6$ ✓

$V_{rms} = \frac{V_m}{\sqrt{2}} = \frac{141.4}{\sqrt{2}}$

$V_{rms} = 99.984$ V

we know that,

$\omega = 2\pi f$

$\therefore f = \frac{\omega}{2\pi} = \frac{628}{2\pi}$

$f = 99.949$ Hz

$\text{form factor} = \frac{\text{Rms value}}{\text{Average value}}$

$= \frac{0.707\ V_m}{0.637\ V_m}$

Form factor $= 1.11$ ✓

Now,

$v = 141.4\sin(628 \times 0.015 - \frac{\pi}{6})$

$v = 71.284$ V

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$\therefore V_m = \sqrt{2} \times 115$

$V_m = 162.634\ V$

We have,

$I_m = \frac{V_m}{R}$

$= \frac{162.634}{70}$

$I_m = 2.32\ A$

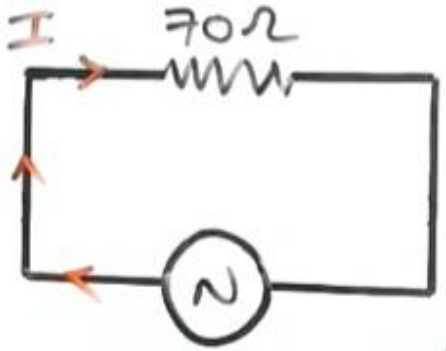
Now, $\omega = 2\pi f$

$= 2\pi \times 50$

$\omega = 314\ rad/sec.$

① ⇒ $V = 162.634 \sin 314t$

$I = 2.32 \sin 314t$



$V = 115V, 50Hz$

std. eqn of V and I:

$V = V_m \sin \omega t$ ①

$I = I_m \sin \omega t$

we know that:

$V_{rms} = \frac{V_m}{\sqrt{2}}$

$\therefore V_m = \sqrt{2} \times V_{rms}$

A 50Hz voltage of 115 V (R.M.S.) is applied across 70Ω resistance. Write the equation of voltage and current.

⇒ Given:

$f = 50Hz$

$V_{rms} = 115\ V$

$R = 70\Omega$

Equation of V = ?

Equation of I = ?

$I = \frac{V}{R}$

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A 50 Hz voltage of 230 V R.M.S. value is applied across a capacitance of 26.5 microFarad. Write the equation for voltage and current in the circuit. Calculate capacitive reactance.

Given:

$f = 50\ Hz$

$V_{rms} = 230\ V$

$C = 26.5\ \mu F$

$= 26.5 \times 10^{-6}\ F$

- Eq. of V = ?
- Eq. of I = ?

$X_C = ?$

$\Rightarrow \omega = 2\pi f$

$= 2\pi \times 50$

$\omega = 314.159\ rad/sec$

$V_{rms} = \frac{V_m}{\sqrt{2}}$

$\therefore V_m = \sqrt{2}\, V_{rms}$

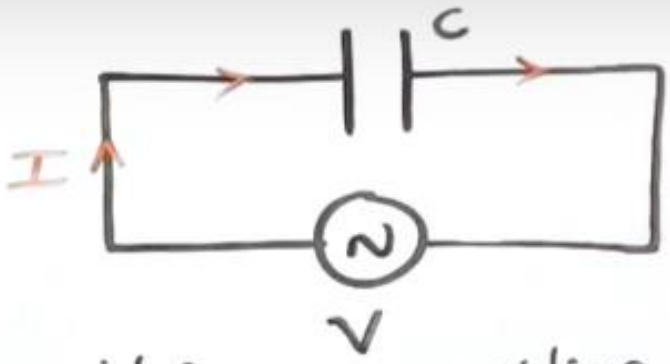
$= \sqrt{2} \times 230$

$V_m = 325.269\ V$

$\therefore$ Eqn of voltage:

$V = V_m \sin \omega t$

$V = 325.269 \sin 314.159t$



$I_m = \frac{V_m}{X_C}$

Since it is a capacitive ckt.

$\therefore$ Current leads the voltage by 90°.

$\therefore \phi = 90°$

$X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi \times 50 \times 26.5 \times 10^{-6}}$

$X_C = 120.116\ \Omega$

Now, $I_m = \frac{V_m}{X_C} = \frac{325.269}{120.116}$

$I_m = 2.707\ A$

$V = V_m \sin \omega t$

$I = I_m \sin(\omega t \pm \phi)$

$\therefore$ Eqn of Current:

$I = I_m \sin(\omega t \pm \phi)$

$I = 2.707 \sin(314.159t + 90)$

Activate Windows

A 50Hz voltage of $120V_{(rms)}$ is applied across 0.1H inductance. Write the equation for voltage and current. Also calculate inductive reactance

Given:

$f = 50Hz$

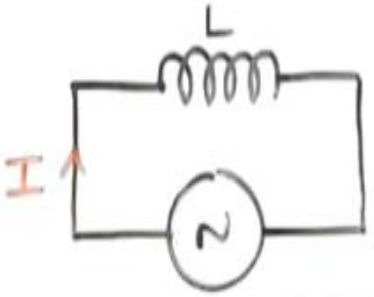
$V_{rms} = 120V$

$L = 0.1\ H$

Eqn of V = ?

Eqn of I = ?

X_L = ?



V = 120V, 50Hz

Since it is an inductive ckt.

∴ Current I lags the voltage V by an angle of 90°.

$\phi = -90°$

Now, $\omega = 2\pi f = 2\pi \times 50$

$\omega = 314.159$ rad/sec

$V = V_m \sin \omega t$

$I = I_m \sin(\omega t \pm \phi)$

we have, $V_{rms} = \frac{V_m}{\sqrt{2}}$

$\therefore V_m = \sqrt{2} \times V_{rms}$

$\therefore V_m = \sqrt{2} \times 120$, $V_m = 169.705\ V$

Eqn of AC voltage, $V = V_m \sin \omega t$

$\therefore V = 169.705 \sin 314t$

Now, $I_m = \frac{V_m}{X_L}$, $X_L = \omega L$

$= 2\pi f L$

$I_m = \frac{169.705}{31.415}$

$= 2\pi \times 50 \times 0.1$

$I_m = 5.402\ A$

$X_L = 31.415\ \Omega$

∴ Eqn of A.C. Current,

$I = I_m \sin(\omega t \pm \phi)$

$I = 5.402 \sin(314t - 90)$

Activate Windows

A coil having 20Ω resistance and 0.2 H inductance is connected across 230V, 50Hz A.C. supply.

Calculate: i) Impedance
ii) Current
iii) Power factor
iv) Power absorbed by the coil.

Given:

R-L ckt.

$R = 20\Omega$

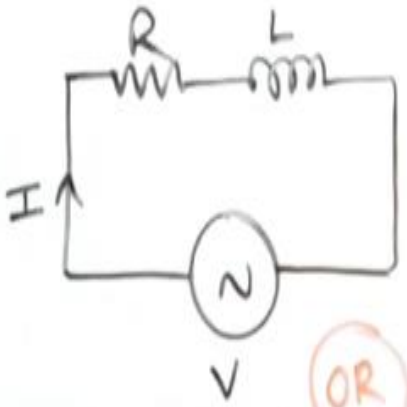
$L = 0.2H$

$V = 230V$

$f = 50Hz$

To find:

- Z = ?
- I = ?
- $\cos\phi$ = ?
- P = ?



$X_L = 2\pi f L$

$= 2\pi \times 50 \times 0.2$

$X_L = 62.83\Omega$

For R-L ckt.

$Z = R + jX_L$

$= 20 + j62.83$

$Z = (65.936 \angle 72.342)\Omega$

$Z = \sqrt{R^2 + X_L^2}$

$= \sqrt{20^2 + 62.83^2}$

OR

$Z = 65.936\Omega$ ✓

$\phi = \tan^{-1}\left(\frac{X_L}{R}\right)$

$= \tan^{-1}\left(\frac{62.83}{20}\right)$

$\phi = 72.342°$ ✓

$I = \frac{V}{Z} = \frac{230}{65.936} = 3.488\ A$

OR

$I = \frac{V}{Z} = \frac{230}{65.936 \angle 72.342}$

$= 3.488 \angle -72.342\ A$

Power factor $= \cos\phi$

$= \cos 72.342$

$= 0.303$ (lagging)

$Power = I^2R = 3.488^2 \times 20$

OR

$= 243.322\ W$

$P = |V||I| \cos\phi = 230 \times 3.488 \cos 72$

$= 243.347\ W$

A series R-L circuit consist of 10Ω resistance and 0·12 H inductance. This series circuit is connected to 150V, 50Hz A.C. supply. Calculate voltage across individual elements.

Given:

R-L circuit

$R = 10\Omega$

$L = 0{\cdot}12\ H$

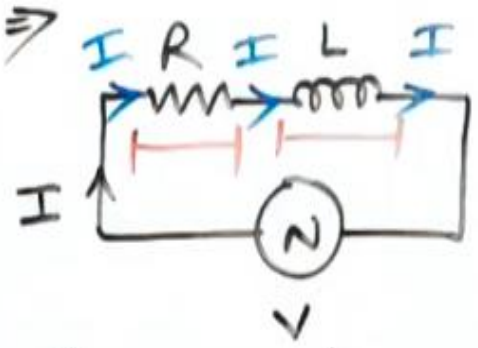
$V = 150\ V$

$f = 50\ Hz$

✓ $V_R = ?$

✓ $V_L = ?$

⇒



$X_L = 2\pi f L$

$= 2\pi \times 50 \times 0{\cdot}12$

$X_L = 37{\cdot}69\Omega$

For R-L ckt.

$Z = \sqrt{R^2 + X_L^2}$

$= \sqrt{10^2 + 37{\cdot}69^2}$

$= 38{\cdot}99$

$Z \simeq 39\Omega$ ✓

$I = \frac{V}{Z}$

$= \frac{150}{39}$

$I = 3{\cdot}84\ A$

$\therefore V_R = I \times R$

$= 3{\cdot}84 \times 10$

$V_R = 38{\cdot}4\ V$ ✓

Now, $V_L = I \times X_L$

$= 3{\cdot}84 \times 37{\cdot}69$

$V_L = 144{\cdot}72\ V$ ✓

$V_R = I \times R$

$V_L = I \times X_L$

$I = \frac{V}{Z}$

Activate Windows
Go to Settings to activate Windows.

A resistor of 20 ohm and an inductance of 0.1 H are connected in series and a 100 V, 50 Hz supply is given to this combination. Find 1) Inductive reactance 2) Impedance 3) Current 4) Power factor 5) Phase difference between v and I 6) Power consumed 7) Voltage across each element.

Given:

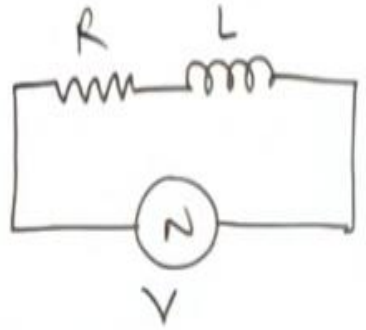
R-L ckt

$R = 20\,\Omega$

$L = 0.1\ H$

$V = 100\ V$

$f = 50\ Hz$



To Find:

X_L = ? Z = ?

I = ? cos ∅ = ?

∅ = ? P = ?

V_R = ? V_L = ?

i) $X_L = 2\pi f L$

$= 2\pi \times 50 \times 0.1$

$X_L = 31.415\,\Omega$

ii) For R-L ckt.

$Z = \sqrt{R^2 + X_L^2}$

$= \sqrt{20^2 + 31.415^2}$

$Z = 37.241\,\Omega$

$Z = R + jX_L$

$= 20 + j31.415$

$Z = 37.24 \angle 57.517\,\Omega$

$\phi = \tan^{-1}\left(\frac{X_L}{R}\right)$

$= \tan^{-1}\left(\frac{31.415}{20}\right)$

$\phi = 57.517^\circ$

Power factor, $\cos\phi = \cos 57.517$

p.f. = 0.537 (lag)

Power factor $= \frac{R}{Z}$

$= \frac{20}{37.241}$

p.f. = 0.537 (lag)

Current, $I = \frac{V}{Z}$

$I = \frac{100}{37.241 \angle 57.517}$

$I = 2.685 \angle -57.517\ A$

Power, $P = |V||I| \cos\phi$

$P = 100 \times 2.685 \times 0.537$

$P = 144.184\ W$

$P = I^2 R = 2.685^2 \times 20$

$P = 144.184\ W$

$V_R = IR$

$= 2.685 \times 20$

$V_R = 53.7\ V$

$V_L = I \times X_L$

$= 2.685 \times 31.415$

$V_L = 84.349\ V$

Activate Windows
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A coil of a resistance 10 ohm and an inductance of 0.2 Henry is connected to 100 V, 50 Hz supply. Calculate
1) Reactance of coil,
2) Impedance of coil,
3) Current taken
4) Phase difference between the current and the applied voltage.

Given:

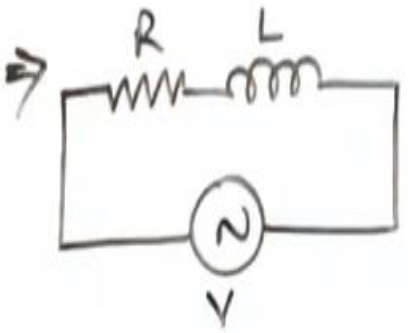
R-L ckt

$R = 10\Omega$

$L = 0.2H$

$V = 100V$

$f = 50Hz$



To find:

i) $X_L = ?$

ii) $Z = ?$

iii) $I = ?$

iv) $\phi = ?$

$X_L = 2\pi f L$

$= 2\pi \times 50 \times 0.2$

$X_L = 62.831\Omega$

$Z = \sqrt{R^2 + X_L^2}$

$Z = \sqrt{10^2 + 62.831^2}$

$Z = 63.621\Omega$

$Z = R + jX_L$

$= 10 + j62.831$

$Z = 63.621 \angle 80.956$

$\phi = \tan^{-1}\left(\frac{X_L}{R}\right) = \tan^{-1}\left(\frac{62.831}{10}\right)$

$\phi = 80.956°$

Current, $I = \frac{V}{Z}$, $I = \frac{V}{Z}$

$= \frac{100}{63.621}$

$= \frac{100}{63.621 \angle 80.956}$

$I = 1.571A$

$I = 1.571 \angle -80.956\ A$

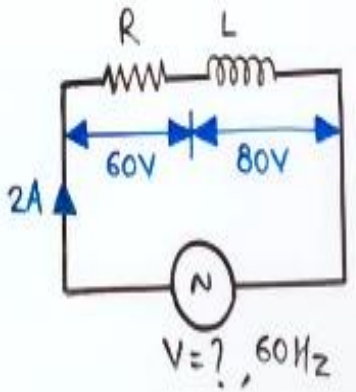
For a circuit shown, find:

i) Values of R and L.

ii) Total Voltage

iii) Impedance

iv) Power factor



➔ **Given:**

R-L ckt

$f = 60Hz$

$I = 2A$

$V_R = 60V$

$V_L = 80V$

To find:

- R=? L=?
- V=?
- Z=?
- $\cos\phi$=?

$\Rightarrow R = \frac{V}{I}$, $X_L = \frac{V_L}{I}$, $P.f. = \frac{R}{Z}$

$= \frac{60}{2}$ $= \frac{80}{2}$ $= \frac{30}{50}$

$R = 30\Omega$ $X_L = 40\Omega$ $P.f. = 0.6$

$X_L = 2\pi f L$

$\therefore L = \frac{X_L}{2\pi f} = \frac{40}{2\pi \times 60}$

$L = 0.106\ H$

Voltage, $V = \sqrt{V_R^2 + V_L^2}$

$V = \sqrt{60^2 + 80^2}$

$V = 100V$

Impedance, $Z = \frac{V}{I} = \frac{100}{2} = 50\Omega$

$Z = 50\Omega$

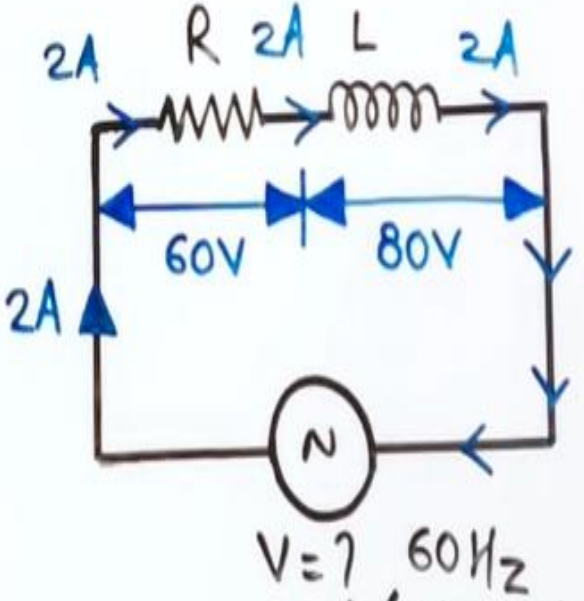
For a circuit shown, find:

i) Values of R and L.

ii) Total Voltage

iii) Impedance

iv) Power factor



Activate Windows
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An a.c. circuit draws a current of 0.5 A at 0.707 p.f. lags from 230 V, 50 Hz a.c. supply. Find an impedance, resistance inductive reactance, power in the circuit.

Given:

R-L ckt.

$I = 0.5A$

p.f. $= 0.707$ (lags)

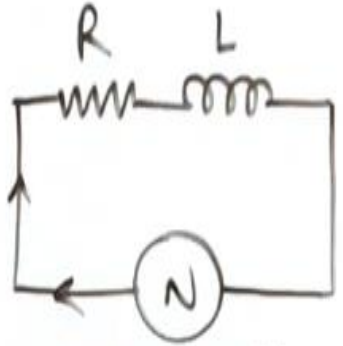
$V = 230V$

$f = 50Hz$

To find:

- $Z = ?$
- $R = ?$
- $X_L = ?$
- $P = ?$

⇒ **Sol^n:**



230 V, 50 Hz

1st method:

$$Z = \frac{V}{I} = \frac{230}{0.5}$$

$Z = 460\Omega$

$R = Z\cos\phi$

$= 460 \times 0.707$

$R = 325.22\Omega$

$X_L = Z\sin\phi$

$X_L = 460\sin 45^\circ$

$X_L = 460 \times 0.707$

$X_L = 325.22\Omega$

Power,

$P = I^2 R$

$= 0.5^2 \times 325.22$

$P = 81.305\,W$

$\cos\phi = 0.707$

$\phi = \cos^{-1}(0.707)$

$\phi = 45^\circ$

2nd method:

$\cos\phi = 0.707$, $\phi = 45^\circ$

$I = 0.5\angle -45\ A$

$$Z = \frac{V}{I} = \frac{230}{0.5\angle -45}$$

$Z = (325.269 + j325.269)\Omega$

$Z = (460\angle 45)\Omega$

$Z = R + jX_L$

$R = 325.269\Omega$

$X_L = 325.269\Omega$

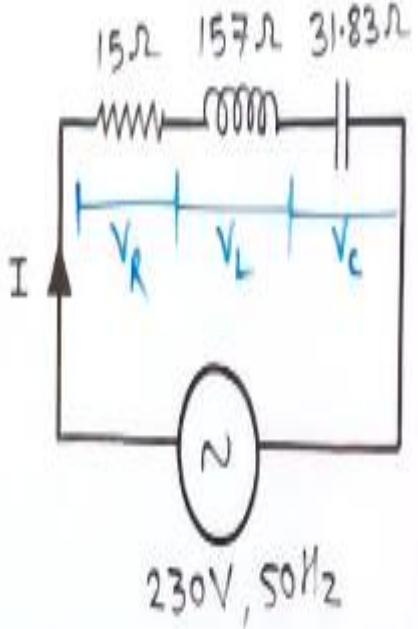
Power,

$P = |V||I|\cos\phi$

$= 230 \times 0.5 \times 0.707$

$P = 81.305\,W$

Activate Windows



Find the current in the given circuit and voltage drop across R, L, C.

Given:

R-L-C ckt.

$R = 15\Omega$

$X_L = 157\Omega$

$X_C = 31.83\Omega$

$V = 230V$

$f = 50Hz$

To Find:

- $I = ?$
- $V_R = ?$
- $V_L = ?$
- $V_C = ?$

$$\Rightarrow Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$= \sqrt{15^2 + (157 - 31.83)^2}$$

$$Z = 126.06\Omega$$

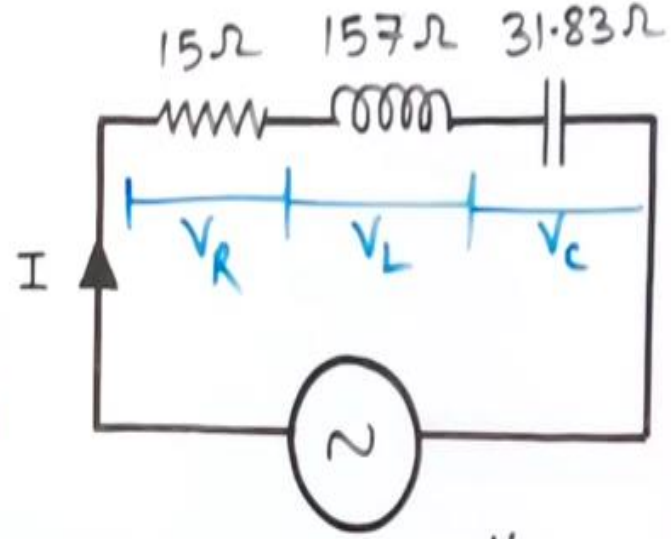
$$I = \frac{V}{Z} = \frac{230}{126.06}$$

$$I = 1.82\ A$$

$$V_R = IR$$

$$= 1.82 \times 15$$

$$V_R = 27.3\ V$$



$$V_L = I \times X_L$$

$$= 1.82 \times 157$$

$$V_L = 285.74\ V$$

$$V_C = I \times X_C$$

$$= 1.82 \times 31.83$$

$$V_C = 57.93\ V$$

(OR)

$$Z = R + j(X_L - X_C)$$

$$= 15 + j(157 - 31.83)$$

$$Z = (15 + j125.17)\Omega$$

$$Z = (126.06 \angle 83.166)\Omega$$

$$I = \frac{V}{Z} = \frac{230}{126.06 \angle 83.166} = (1.82 \angle -83.166)\ A$$

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A coil of resistance 10 ohm and inductance 0.1 Henry is connected in series with capacitor of 150 microfarad across 200 Volt, 50 Hz supply. Calculate
1) Inductive reactance
2) Capacitive reactance
3) Impedance
4) Current.

Given:

R-L-C ckt

$R = 10\,\Omega$

$L = 0.1\,H$

$C = 150\,\mu F = 150 \times 10^{-6}\,F$

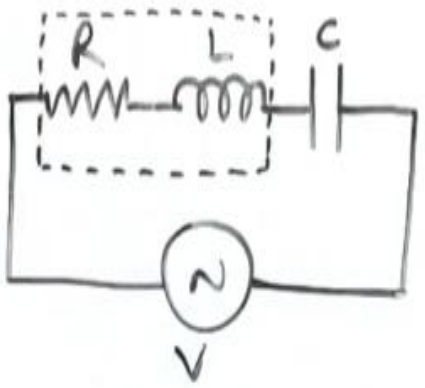
$V = 200\,V$

$f = 50\,Hz$

To find:

$X_L = ?$ $X_C = ?$

$Z = ?$ $I = ?$



$X_L = 2\pi f L$

$= 2\pi \times 50 \times 0.1$

$X_L = 31.41\,\Omega$

$X_C = \frac{1}{2\pi f C}$

$= \frac{1}{2\pi \times 50 \times 150 \times 10^{-6}}$

$X_C = 63.66\,\Omega$

For R-L-C ckt.

$Z = \sqrt{R^2 + (X_L - X_C)^2}$

$= \sqrt{10^2 + (31.41 - 63.66)^2}$

$Z = 33.75\,\Omega$ ✓

$I = \frac{V}{Z}$

$= \frac{200}{33.75}$

$I = 5.92\,A$

2nd method

$Z = R + j(X_L - X_C)$

$= 10 + j(31.41 - 63.66)$

$Z = (10 - j32.25)\,\Omega$

$Z = (33.764 \angle -72.772)\,\Omega$

$I = \frac{V}{Z}$

$= \frac{200}{33.764 \angle -72.772}$

$I = (5.923 \angle 72.772)\,A$

Activate Windows
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A resistance of 20 ohm, inductance of 0.2 H and capacitance of 150 microF. are connected in series to 230 V, 50 Hz AC supply. Find 1) Impedance of the circuit 2) Power factor 3) Active power 4) Reactive power

Given:

R-L-C ckt.

$R = 20\,\Omega$

$L = 0.2\,H$

$C = 150\,\mu F = 150 \times 10^{-6}\,F$

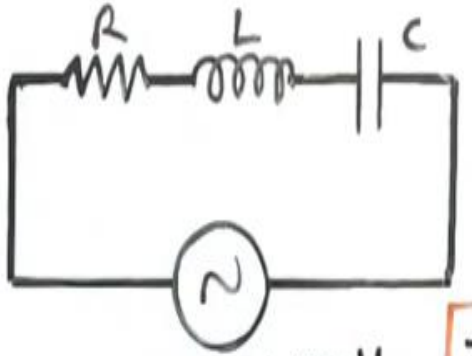
$V = 230\,V$

$f = 50\,Hz$

To Find:

- $Z = ?$
- $\cos\phi = ?$
- $P = ?$ $Q = ?$

$\Rightarrow$



230V, 50Hz

$X_L = 2\pi f L$

$= 2\pi \times 50 \times 0.2$

$X_L = 62.831\,\Omega$

$X_C = \frac{1}{2\pi f C}$

$= \frac{1}{2\pi \times 50 \times 150 \times 10^{-6}}$

$X_C = 21.22\,\Omega$

For R-L-C ckt.

$Z = R + j(X_L - X_C)$

$= 20 + j(62.831 - 21.22)$

$Z = (20 + j41.611)\,\Omega$

$Z = 46.167 \angle 64.329\,\Omega$

$I = \frac{V}{Z}$

$= \frac{230}{46.167 \angle 64.329}$

$I = 4.981 \angle -64.329\,A$

Power factor $= \cos\phi$

$= \cos 64.329$

P.f. $= 0.433$ (lagging)

Active Power,

$P = |V||I| \cos\phi$

$= 230 \times 4.981 \times 0.433$

$\therefore P = 496.057\,W$

Reactive Power,

$Q = |V||I| \sin\phi$

$= 230 \times 4.981 \sin 64.329$

$\therefore Q = 1032.552\,VAR$

Activate Windows
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Find the active and reactive components of the current taken by a series circuit consisting of a coil of inductance 0.1H, resistance 8ohm and capacitor of 120 micro Farad connected to a 240V, 50 Hz supply.

Given:

R-L-C ckt

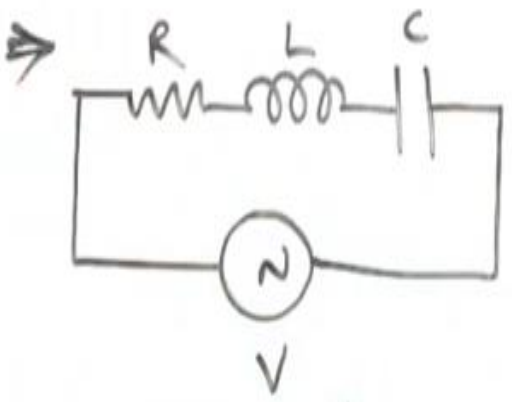
$R = 8\Omega$

$L = 0.1\,H$

$C = 120 \times 10^{-6}\,F$

$V = 240\,V$

$f = 50\,Hz$



$X_L = 2\pi f L$

$= 2\pi \times 50 \times 0.1$

$X_L = 31.415\,\Omega$

Active component of I = ?

Reactive component of I = ?

$X_C = \frac{1}{2\pi f C}$

$= \frac{1}{2\pi \times 50 \times 120 \times 10^{-6}}$

$X_C = 26.525\,\Omega$

For R-L-C ckt.

$Z = R + j(X_L - X_C)$

$= 8 + j(31.415 - 26.525)$

$Z = (8 + j4.89)\,\Omega$

Now, $I = \frac{V}{Z} = \frac{240}{8 + j4.89}$

$I = 25.596 \angle -31.435\,A$

$I = 21.839 - j13.349\,A$

$\therefore$ Active Component of I = 21.839

Reactive Component of I = 13.349

A resistance of 10ohm is connected in series with an inductance of 0.05 H and a capacitance of 300 microfarad to 100V, AC supply. Calculate the magnitude & phase angle of current when frequency is
1) 25 cycles per second
2) 50 cycles per second.

Given:

R-L-C ckt

$R = 10\,\Omega$

$L = 0.05\,H$

$C = 300\,\mu F$

$= 300 \times 10^{-6}\,F$

$V = 100\,V$

$I = ?\quad \phi = ?$

a) At $f = 25\,Hz$

b) At $f = 50\,Hz$

⇒ a) When $f = 25\,Hz$

$X_L = 2\pi f L$

$= 2\pi \times 25 \times 0.05$

$X_L = 7.853\,\Omega$

$X_C = \frac{1}{2\pi f C}$

$= \frac{1}{2\pi \times 25 \times 300 \times 10^{-6}}$

$X_C = 21.22\,\Omega$

For RLC ckt.

$Z = R + j(X_L - X_C)$

$= 10 + j(7.853 - 21.22)$

$Z = 10 - j13.367\,\Omega$

Now, Current $I = \frac{V}{Z}$

$I = \frac{100}{10 - j13.367}$

$I = 5.99 \angle 53.199\,A$

magnitude, phase Angle

$I = \frac{V}{Z}$

$Z = R + j(X_L - X_C)$

$Z = \sqrt{R^2 + (X_L - X_C)^2}$

$I = \frac{V}{Z}$

$Z = R + j(X_L - X_C)$

$Z = \sqrt{R^2 + (X_L - X_C)^2}$

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Given:

R-L-C ckt

$R = 10\ \Omega$

$L = 0.05\ H$

$C = 300\ \mu F$

$= 300 \times 10^{-6}\ F$

$V = 100\ V$

$I = ?\quad \phi = ?$

a) At $f = 25\ Hz$

b) At $f = 50\ Hz$

⇒ b) When $f = 50\ Hz$

$X_L = 2\pi f L$

$= 2\pi \times 50 \times 0.05$

$X_L = 15.707\ \Omega$

$X_C = \frac{1}{2\pi f C}$

$= \frac{1}{2\pi \times 50 \times 300 \times 10^{-6}}$

$X_C = 10.61\ \Omega$

for RLC ckt.

$Z = R + j(X_L - X_C)$

$= 10 + j(15.707 - 10.61)$

$Z = 10 + j5.097\ \Omega$

Now, Current $I = \frac{V}{Z}$

$I = \frac{100}{10 + j5.097}$

$I = 8.909 \angle -27.007\ A$

magnitude, phase Angle

∴ Current, $I = 8.909\ A$

phase Angle $\phi = 27.007°$

Activate Windows
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A voltage e=150sin314t is applied across a series RLC circuit, R=40ohm, L= 0.3 H, C = 40 microF, Compute : 1) RMS value of steady state current 2) Power supplied by the source 3) Time expression of current.

given:

RLC ckt

$e = 150\sin 314t$

$R = 40\,\Omega$

$L = 0.3\,H$

$C = 40\times10^{-6}\,F$

→ $I_{rms} = ?$ ✓

→ $P = ?$

→ eqn. of Current = ?

Eqn of Current. ⇒ Given eqn of voltage, $e = 150\sin 314t$

$i(t) = I_m\sin(\omega t \pm \phi)$ ——①

compair it with $V = V_m\sin\omega t$

$\therefore V_m = 150\,V$

$\omega = 314$ rad/sec

$I_{rms} = \frac{I_m}{\sqrt{2}}$

$\therefore I_m = \sqrt{2}\,I_{rms}$

$= \sqrt{2}\times(2.491)$

$I_m = 3.522\,A$

$\therefore V_{rms} = \frac{V_m}{\sqrt{2}}$

$= \frac{150}{\sqrt{2}}$

$V_{rms} = 106.066\,V$

① ⇒ $i(t) = 3.522\sin(314t - 20.03)$

$X_L = \omega L$

$= 314\times0.3$

$X_L = 94.2\,\Omega$

$X_C = \frac{1}{\omega C}$

$= \frac{1}{314\times40\times10^{-6}}$

$X_C = 79.617\,\Omega$

For RLC ckt

$Z = R + j(X_L - X_C)$

$= 40 + j(94.2 - 79.617)$

$Z = (40 + j14.583)\,\Omega$

Current, $I_{rms} = \frac{V_{rms}}{Z}$

$= \frac{106.066}{40 + j14.583}$

$I_{rms} = 2.491\angle -20.03\,A$

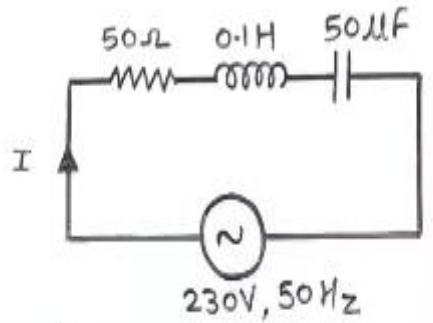
Power,

$P = |V||I|\cos\phi$

$= 106.066\times2.491\cos 20.03$

$\therefore P = 248.229\,W$ ✓

Activate Windows



For the circuit shown:
Calculate the following
i) Impedance
ii) Power consumed by circuit
iii) Power factor & its nature.

Given:

R-L-C ckt.

$R = 50\,\Omega$

$L = 0.1\,H$

$C = 50\,\mu F = 50 \times 10^{-6}\,F$

$V = 230\,V$

$f = 50\,Hz$

$Z = ?$

$P = ?$

$\cos\phi = ?$

$\Rightarrow X_L = 2\pi f L$

$= 2\pi \times 50 \times 0.1$

$X_L = 31.415\,\Omega$

$X_C = \frac{1}{2\pi f C}$

$= \frac{1}{2\pi \times 50 \times 50 \times 10^{-6}}$

$X_C = 63.661\,\Omega$

For R-L-C ckt.

$Z = R + j(X_L - X_C)$

$= 50 + j(31.415 - 63.661)$

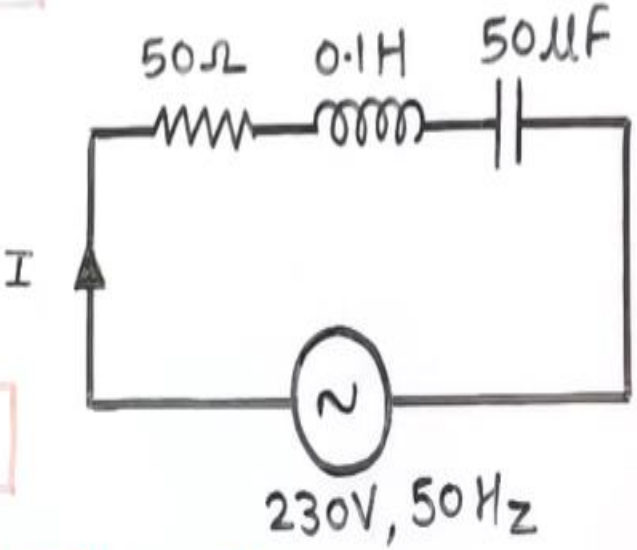
$Z = (50 - j32.246)\,\Omega$ ✓

$Z = 59.496 \angle -32.818\,\Omega$ ✓

Current, $I = \frac{V}{Z}$

$= \frac{230}{59.496 \angle -32.818}$

$I = 3.865 \angle 32.818\,A$



Power, $P = |V|\,|I| \cos\phi$

$= 230 \times 3.865 \cos 32.818$

$P = 747.07\,W$ ✓

Power factor, $P.f. = \cos\phi$

$= \cos 32.818$

$P.f. = 0.84$ (leading) ✓

Activate Windows
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A sinusoidal 50 c/s current of maximum value of 100 A flows throug a capacitor of 318 microF capacitance. Calculate : 1) The expression for instantaneous current.
2) Reactance of the capacitor.
3) Equation of applied EMF.
4) RMS values of applied EMF and current.

$$I_{rms} = \frac{I_m}{\sqrt{2}} = \frac{100}{\sqrt{2}}$$

$$I_{rms} = 70.71\ A$$

Given:

$f = 50\ Hz$

$I_m = 100\ A$

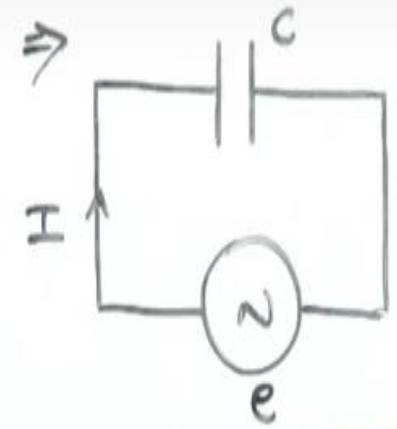
$C = 318\ \mu F = 318 \times 10^{-6}\ F$

✓ i) eqn of current

✓ ii) X_C

✓ iii) eqn of e.m.f.

iv) $I_{rms} = ?$, $e_{rms} = ?$



$$I = I_m \sin(\omega t \pm \phi) \quad \text{— (1)}$$

$\omega = 2\pi f$

$= 2\pi \times 50$

$\omega = 314.159\ rad/sec$

$\omega = 314\ rad/sec.$

Since it is a pure capacitive circuit, $\phi = 90°$

(1) ⇒ $I = 100 \sin(314t + 90)$

$$X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi \times 50 \times 318 \times 10^{-6}}$$

$X_C = 10\ \Omega$

Equation of e.m.f.

$e = e_m \sin \omega t$ — (2)

$I_m = \frac{e_m}{X_C}$, $e_m = I_m \times X_C$

$= 100 \times 10$

$e_m = 1000\ V$

(2) ⇒ $e = 1000 \sin 314t$

$e_{rms} = \frac{e_m}{\sqrt{2}} = \frac{1000}{\sqrt{2}}$, $e_{rms} =$

Activate Windows

A capacitor and resistor are connected in series across a 220 V, 50 Hz supply. The circuit draws a current of 2.5 A. If power lost in the circuit is 250 W. Find values of resistance and capacitance.

Given:

R-C ckt.

$V = 220V$

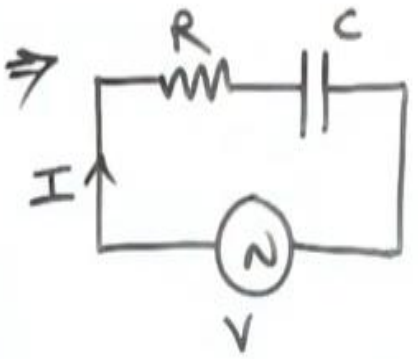
$f = 50Hz$

$I = 2.5A$

$P = 250W$

$R = ?$

$C = ?$



Power, $P = I^2R$

$\therefore R = \frac{P}{I^2}$

$= \frac{250}{2.5^2}$

$R = 40\Omega$

Current, $I = \frac{V}{Z}$

$\therefore Z = \frac{V}{I} = \frac{220}{2.5}$

$Z = 88\Omega$

For R-C ckt.

$Z = \sqrt{R^2 + X_C^2}$

$\therefore X_C = \sqrt{Z^2 - R^2}$

$X_C = \sqrt{88^2 - 40^2}$

$X_C = 78.38\Omega$

we have, $X_C = \frac{1}{2\pi fC}$

$C = \frac{1}{2\pi f X_C}$

$= \frac{1}{2\pi \times 50 \times 78.38}$

$C = 4.061 \times 10^{-5} F$

$P = I^2R$

$X_C = \frac{1}{2\pi fC}$

$Z = \sqrt{R^2 + X_C^2}$

$C = 40.61 \times 10^{-6} F$

$C = 40.61 \mu F$

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Waveform of the voltage and current in a circuit are given by the equations:

$V = 150 \sin 314t$

$I = 10 \sin\left(314t + \frac{\pi}{4}\right)$

Draw the circuit and vector diag. Calculate frequency of supply and phase angle between voltage and current.

Given:

$V = 150 \sin 314t$

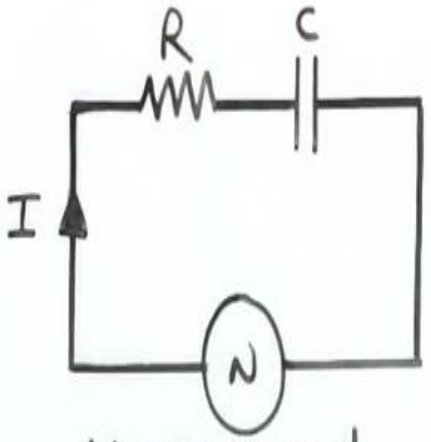
$I = 10 \sin\left(314t + \frac{\pi}{4}\right)$

- ckt diag. = ?
- phasor diag. = ?
- f = ?
- ∅ = ?
- R = ?
- C = ?

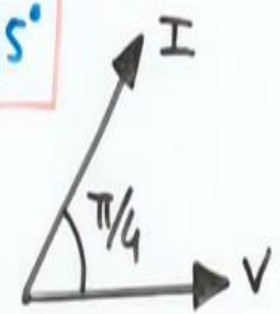
$Z = \frac{V}{I}$

⇒ Since $\frac{\pi}{4}$ is +ve

∴ the given ckt. is R-C ckt.



$\phi = \frac{\pi}{4} = 45°$



Given that: $V = 150 \sin 314t$

compair it with $V = V_m \sin \omega t$

∴ $V_m = 150\,V$, $\omega = 314$ rad/sec

Given that $I = 10 \sin(314t + \pi/4)$

Compair it with $I = I_m \sin(\omega t \pm \phi)$

$I_m = 10\,A$, $\omega = 314$ rad/sec, $\phi = \pi/4$

we know that, $\omega = 2\pi f$

$\therefore f = \frac{\omega}{2\pi}$, $f = \frac{314}{2\pi}$, $f = 49.497\,Hz$

$V_{rms} = \frac{V_m}{\sqrt{2}} = \frac{150}{\sqrt{2}} = 106.066\,V$

$I_{rms} = \frac{I_m}{\sqrt{2}} = \frac{10}{\sqrt{2}} = 7.071\,A$

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$$Z = \frac{V}{I}$$

$$= \frac{106.066}{7.071}$$

$$Z = 15\ \Omega \quad \checkmark$$

Power factor,

$$\cos\phi = \frac{R}{Z}$$

$$\therefore R = Z\cos\phi$$

Radian

$$R = 15\cos\frac{\pi}{4}$$

$$R = 10.6\ \Omega \quad \checkmark$$

for R-C ckt -

$$Z = \sqrt{R^2 + X_C^2}$$

$$\therefore X_C = \sqrt{Z^2 - R^2}$$

$$= \sqrt{15^2 - 10.6^2}$$

$$\therefore X_C = 10.613\ \Omega$$

we have,

$$X_C = \frac{1}{2\pi f C}$$

$$C = \frac{1}{2\pi f X_C}$$

$$= \frac{1}{2\pi \times 49.497 \times 10.613}$$

$$C = 3.029 \times 10^{-4}\ F$$

$$C = 302.9 \times 10^{-6}\ F, \quad C = 302.9\ \mu F$$

A capacitor of capacitance 79.5μF is connected in series with a non-inductive resistance of 30Ω across 100V, 50Hz supply.

Find i) Impedance
ii) Current
iii) Phase angle
iv) Power factor.

Given:

R-C ckt

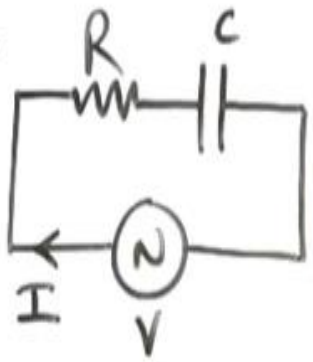
$C = 79.5\mu F$

$= 79.5 \times 10^{-6} F$

$R = 30\Omega$

$V = 100V$

$f = 50Hz$



(lead)

Find: $Z = ?$

$I = ?$

$\phi = ?$

$\cos\phi = ?$

$X_C = \frac{1}{2\pi fC}$

$= \frac{1}{2\pi \times 50 \times 79.5 \times 10^{-6}}$

$X_C = 40.038\Omega$

For R-C ckt.

$Z = R - jX_C$

$= 30 - j40.038$

$Z = (50.03\angle -53.156)\Omega$

Current, $I = \frac{V}{Z} = \frac{100}{50.03\angle -53.156}$

$I = 1.998\angle 53.156\ A$

Phase angle, $\phi = \tan^{-1}\left(\frac{-X_C}{R}\right)$

$= \tan^{-1}\left(\frac{-40.038}{30}\right)$

$\phi = -53.156^\circ$

Power factor, $p.f. = \cos\phi$

$= \cos 53.156$

$p.f. = 0.6$ (leading)

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A choke coil takes a current of 2.5A. When connected across 230V, 50Hz ac supply and consumes 400W. Calculate :

i) Resistance of coil
ii) Power factor
iii) Inductive reactance
iv) Inductance of coil.

Given :

R-L ckt.

$I = 2.5\,A$

$f = 50\,Hz$

$V = 230\,V$

$P = 400\,W$

$R = ?$

$\cos\phi = ?$

$X_L = ?$

$L = ?$

$P = V I \cos\phi$

$\cos\phi = \frac{P}{VI}$

$= $

$\Rightarrow$ Power

$P = I^2 R$

$400 = 2.5^2 \times R$

$\therefore R = 64\,\Omega$

Impedance,

$Z = \frac{V}{I}$

$= \frac{230}{2.5}$

$Z = 92\,\Omega$

$\cos\phi = \frac{R}{Z}$

$= \frac{64}{92}$

$\cos\phi = 0.695$ (lagging)

For R-L ckt

$Z = \sqrt{R^2 + X_L^2}$

$\therefore X_L = \sqrt{Z^2 - R^2}$

$X_L = \sqrt{92^2 - 64^2}$

$X_L = 66.09\,\Omega$

we have, $X_L = 2\pi f L$

$\therefore L = \frac{X_L}{2\pi f}$

$= \frac{66.09}{2\pi \times 50}$

$L = 0.21\,H$

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4. A current of 5A flows through a series resistance and inductance circuit when a 250V, 50 Hz supply is connected. If voltage across resistance is 100V, then calculate :

(i) resistance

(ii) voltage across inductance

(iii) inductance

(iv) power factor

(v) active power

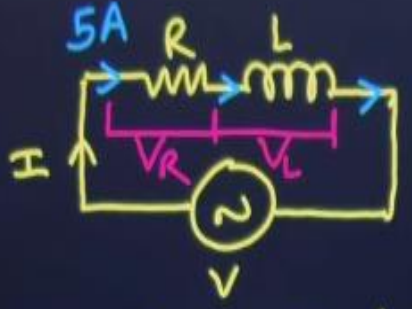
(vi) reactive power

Given: R-L ckt

- $I = 5A$
- $V = 250V$
- $f = 50Hz$
- $V_R = 100V$

To find:

- $R = ?$
- $V_L = ?$
- $L = ?$
- $\cos\phi = ?$
- $P = ?$ $Q = ?$



We know that

$V_R = IR$

$100 = 5 \times R$

$\therefore R = \frac{100}{5}$

$R = 20\Omega$

Now, $I = \frac{V}{Z}$

$\therefore Z = \frac{V}{I} = \frac{250}{5}$

$Z = 50\Omega$

Active Power, $P = V \times I \times \cos\phi$

$= 250 \times 5 \times 0.4$

$P = 500W$

for RL ckt

$Z = \sqrt{R^2 + X_L^2}$

$\therefore X_L = \sqrt{Z^2 - R^2}$

$= \sqrt{50^2 - 20^2}$

$X_L = 45.825\Omega$

$X_L = 2\pi f L$

$\therefore L = \frac{X_L}{2\pi f}$

$= \frac{45.825}{2\pi \times 50}$

$L = 0.145H$

Now, $V_L = I \times X_L$

$= 5(45.825)$

$V_L = 229.125V$

Now, power factor,

$\cos\phi = \frac{R}{Z}$

$\cos\phi = \frac{20}{50}$

$\cos\phi = 0.4$ (Lagging)

Reactive Power,

$Q = V \times I \times \sin\phi$

$= 250 \times 5 \times \sin 66.421$

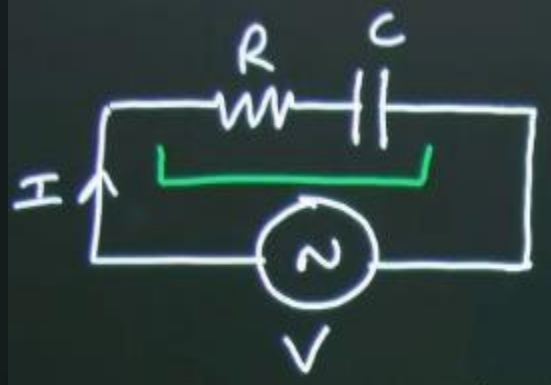
$Q = 1145.636 VAR$

$\cos\phi = 0.4$

$\phi = 66.421$

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A capacitive reactance of 4Ω is connected in series with a resistance of 5Ω. The series circuit is connected across a 200V, 50Hz supply. Find
i) Capacitance
ii) Impedance.
iii) Power factor.



Given:

R-C ckt

$X_C = 4\Omega$

$R = 5\Omega$

$V = 200\,V$

$f = 50\,Hz$

- $C = ?$
- $Z = ?$
- $\cos\phi = ?$

we know that

$$X_C = \frac{1}{2\pi f C}$$

$$C = \frac{1}{2\pi f X_C}$$

$$C = \frac{1}{2\pi \times 50 \times 4}$$

$$C = 7.957 \times 10^{-4}\ F$$

$$C = 795.7 \times 10^{-6}\ F$$

$$C = 795.7\ \mu F$$

for RC ckt

$$Z = \sqrt{R^2 + X_C^2}$$

$$= \sqrt{5^2 + 4^2}$$

$$Z = 6.403\ \Omega$$

$$Z = R - jX_C$$

$$Z = 5 - j4$$

$$Z = 6.403 \angle -38.659\ \Omega$$

Power factor, $P.f. = \cos\phi$

$= \cos 38.659$

$P.f. = 0.78$ (leading)

Power factor, $\cos\phi = \frac{R}{Z}$

$= \frac{5}{6.403}$

$\cos\phi = 0.78$ (leading)

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An impedance of (10-j50) ohm is connected across 220V, 50 Hz supply. Calculate current taken, power factor and power consumed.

Given:

$Z = 10 - j50\ \Omega$

$V = 220V$

$f = 50Hz$

✓ $I = ?$

✓ $\cos\phi = ?$

✓ $P = ?$

⇒ we know that,

$$I = \frac{V}{Z}$$

$$= \frac{220}{10 - j50}$$

$$I = 4.314\angle 78.69\ A$$

$$\text{Power factor} = \cos\phi$$

$$= \cos 78.69$$

$$\text{P.F.} = 0.196 \text{ (leading)}$$

Now, Power, $P = |V||I|\cos\phi$

$$= 220 \times 4.314 \cos 78.69$$

$$\therefore P = 186.13\ W$$

we have, $Z = 10 - j50\ \Omega$

$$Z = 50.99\angle -78.69\ \Omega$$

$$I = \frac{V}{Z} = \frac{220}{50.99\angle -78.69}$$

$$I = 4.314\angle 78.69\ A$$

Power factor

$$\cos\phi = \frac{R}{Z}$$

$$= \frac{10}{50.99}$$

$$= 0.196 \text{ (leading)}$$

Power,

$$P = I^2 R$$

$$P = 4.314^2 \times 10$$

$$P = 186.105\ W$$

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A resistance of 10 ohm, inductance of 0.1 H and capacitance of 100 microfarad are connected in series across 100V, 50 Hz A.C. supply.
Find :
1) Current
2) Power factor
3) Power
4) Draw vector diagram.

Given :

R-L-C ckt

$R = 10\,\Omega$

$L = 0.1\,H$

$C = 100\,\mu F$

$= 100 \times 10^{-6}\,F$

$V = 100\,V$

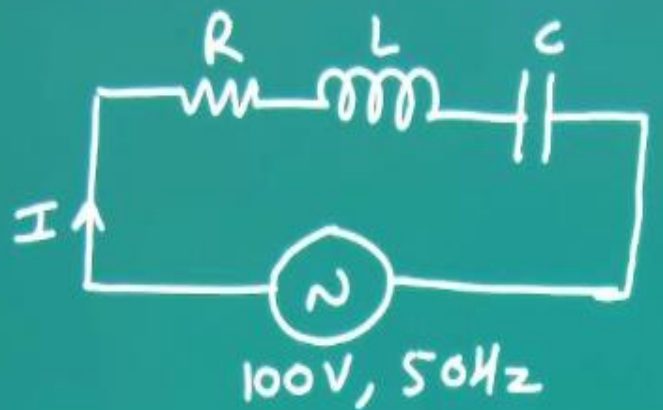
$f = 50\,Hz$

✓ $I = ?$

✓ $\cos\phi = ?$

✓ $P = ?$

Vector diag. = ?



$X_L = 2\pi f L$

$= 2\pi \times 50 \times 0.1$

$X_L = 31.41\,\Omega$

$X_C = \frac{1}{2\pi f C}$

$= \frac{1}{2\pi \times 50 \times 100 \times 10^{-6}}$

$X_C = 31.83\,\Omega$

For R-L-C ckt.

$Z = \sqrt{R^2 + (X_L - X_C)^2}$

$= \sqrt{10^2 + (31.41 - 31.83)^2}$

$Z = 10\,\Omega$

Current, $I = \frac{V}{Z}$

$= \frac{100}{10}$

$I = 10\,A$

Power factor, $\cos\phi = \frac{R}{Z}$

$\cos\phi = \frac{10}{10}$

$\cos\phi = 1$

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$X_L = 2\pi f L$

$= 2\pi \times 50 \times 0.1$

$\boxed{X_L = 31.41\ \Omega}$

$X_C = \frac{1}{2\pi f C}$

$= \frac{1}{2\pi \times 50 \times 100 \times 10^{-6}}$

$\boxed{X_C = 31.83\ \Omega}$

Power, $P = V I \cos\phi$

$= 100 \times 10 \times 1$

$\boxed{P = 1000\ W}$

OR

$P = I^2 R$

$= 10^2 \times 10$

$\boxed{P = 1000\ W}$

For R-L-C ckt.

$Z = \sqrt{R^2 + (X_L - X_C)^2}$

$= \sqrt{10^2 + (31.41 - 31.83)^2}$

$\boxed{Z = 10\ \Omega}$

Current, $I = \frac{V}{Z}$

$= \frac{100}{10}$

$\boxed{I = 10\ A}$

Power factor, $\cos\phi = \frac{R}{Z}$

$\cos\phi = \frac{10}{10}$

$\boxed{\cos\phi = 1}$

$\therefore \boxed{\phi = 0}$



Waveform of the voltage and current in a circuit are given by equations $V = 150 \sin 314t$, $I = 10 \sin (314t - \frac{\pi}{3})$. Draw the circuit and vector diagram. Find the value of resistance and inductance which are connected in series to form the circuit.

Given:

R-L ckt.

$V = 150 \sin 314t$

$I = 10 \sin (314t - \frac{\pi}{3})$

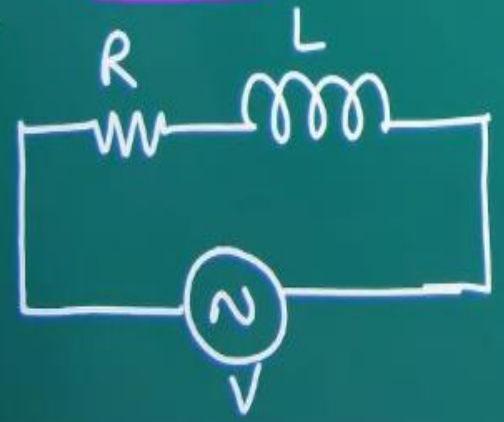
✓ ckt diag. = ?

✓ Vector diag = ?

R = ?

L = ?

⇒



Given that :

$V = 150 \sin 314t$

Compair it with

$v = V_m \sin \omega t$

$\therefore V_m = 150\ V$

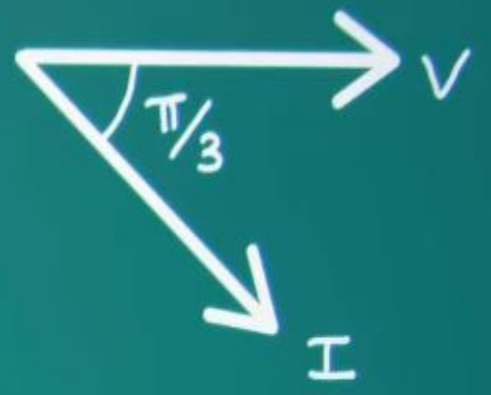
$\omega = 314\ rad/sec.$

And $I = 10 \sin (314t - \frac{\pi}{3})$

compair it with

$I = I_m \sin (\omega t - \phi)$

$\therefore I_m = 10\ A$, $\phi = \frac{\pi}{3}$



Now, $V_{rms} = \frac{V_m}{\sqrt{2}}$

$= \frac{150}{\sqrt{2}}$

$V_{rms} = 106.066\ V$

$I_{rms} = \frac{I_m}{\sqrt{2}}$

$= \frac{10}{\sqrt{2}}$

$I_{rms} = 7.071\ A.$

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Now,

$I_{rms} = 7.071\ A.$

Impedance, $Z = \frac{V}{I}$

$Z = \frac{106.066}{7.071}$

$Z = 15\ \Omega$

we know that,

$\cos\phi = \frac{R}{Z}$

$\therefore R = Z\cos\phi = 15\cos\left(\frac{\pi}{3}\right)$

$R = 7.5\ \Omega$

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$R = 7.5\ \Omega$

For R-L ckt: $Z = \sqrt{R^2 + X_L^2}$

$\therefore X_L = \sqrt{Z^2 - R^2}$

$= \sqrt{15^2 - 7.5^2}$

$X_L = 12.99\ \Omega$

we have,

$X_L = 2\pi f L$

$X_L = \omega L$

$\omega = 2\pi f$

$12.99 = 314 \times L$

$\therefore L = \frac{12.99}{314}$

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A series R-L-C circuit consist of $R = 10\,\Omega$, $L = 0.318\,H$, $C = 63.6\,\mu F$ and $e = 100 \sin 314t$.

Calculate:

i) Expression for $i(t)$

ii) Phase angle between V and I.

iii) Power factor

iv) Active power consumed.

v) Draw the phasor diagram.

Given:

Series R-L-C ckt

$R = 10\,\Omega$

$L = 0.318\,H$

$C = 63.6\,\mu F$

$= 63.6 \times 10^{-6}\,F$

$e = 100 \sin 314t$

To Find:

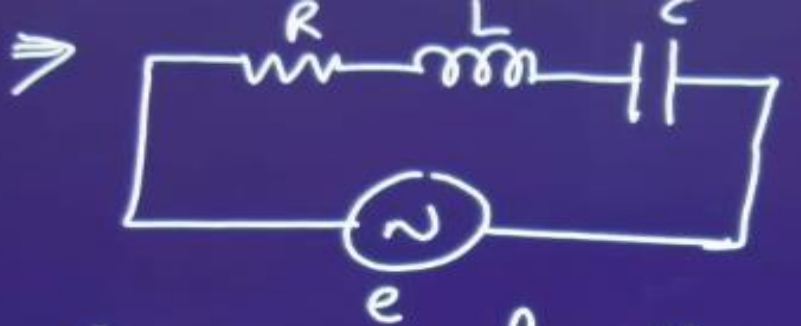
i) eqn of $i(t)$

ii) ϕ

iii) $\cos\phi$

iv) P

v) Phasor diag.



Given eqn of voltage:

$e = 100 \sin 314t$

compair it with:

$v = V_m \sin \omega t$

$\therefore V_m = 100\,V$, $\omega = 314$

$\therefore V_{rms} = \frac{V_m}{\sqrt{2}} = \frac{100}{\sqrt{2}}$

$\therefore V_{rms} = 70.71\,V$ ✓

$Z = R + j(X_L - X_C)$

$Z = \sqrt{R^2 + (X_L - X_C)^2}$

$X_L = \omega L$

$= 314 \times 0.318$

$X_L = 99.852\ \Omega$

$X_C = \frac{1}{\omega C}$

$= \frac{1}{314 \times 63.6 \times 10^{-6}}$

$X_C = 50.074\ \Omega$

∴ For R-L-C ckt.

$Z = R + j(X_L - X_C)$

$= 10 + j(99.852 - 50.074)$

$Z = 10 + j49.778\ \Omega$

OR

$Z = 50.772 \angle 78.64\ \Omega$

∴ Current, $I = \frac{V}{Z}$

$I = \frac{70.71}{50.772 \angle 78.64}$

$I = 1.392 \angle -78.64\ A$

$I_{rms} = \frac{I_m}{\sqrt{2}}$

∴ $I_m = \sqrt{2}\, I_{rms}$

$I_m = \sqrt{2} \times 1.392$

$I_m = 1.968\ A$

Now,

Eqn of Current i(t),

$i(t) = I_m \sin(\omega t \pm \phi)$

$i(t) = 1.968 \sin(314t - 78.64)$

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ii) Phase angle ϕ bet^n V and I,

$\phi = 78.64^\circ$

iii) Power factor = $\cos\phi$

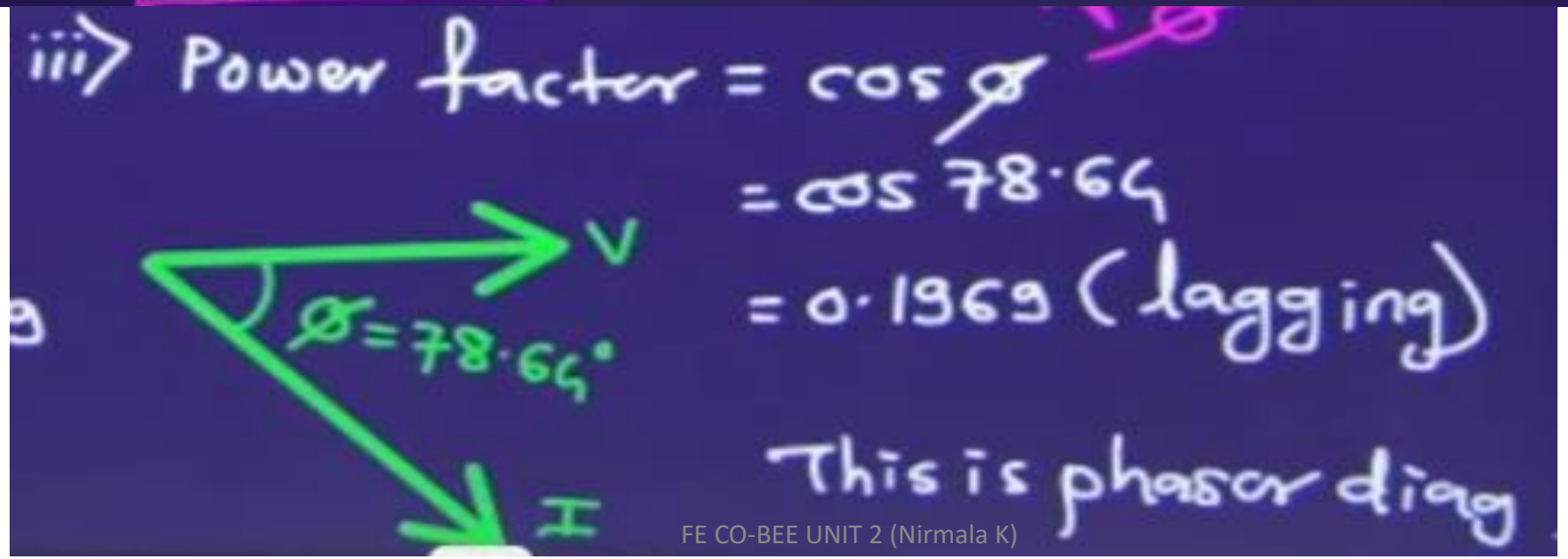
$= \cos 78.64$

$= 0.1969$ (lagging)

iv) Power, $P = |V||I| \cos\phi$

$= 70.71 \times 1.392 \times 0.1969$

$P = 19.38$ W ✓



An inductive coil of resistance 32Ω and reactance 15.7Ω is connected in series with a capacitive reactance 79.5Ω. The circuit as a whole is connected across 500 V A.C. supply. Determine:

1) Current
2) Power factor
3) Voltage across inductive coil
4) Total power absorbed.

Given:

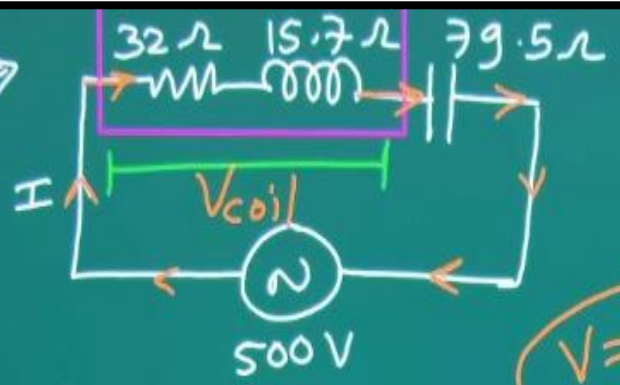
R-L-C ckt

$R = 32\ \Omega$

$X_L = 15.7\ \Omega$

$X_C = 79.5\ \Omega$

$V = 500\ V$



To Find:

- $I = ?$
- $\cos\phi = ?$
- $V_{coil} = ?$
- $P = ?$

For R-L-C ckt

$Z = R + j(X_L - X_C)$

$Z = 32 + j(15.7 - 79.5)$

$Z = (32 - j63.8)\ \Omega$

Now, Current, $I = \frac{V}{Z}$

$I = \frac{500}{32 - j63.8}$

$I = 7.005 \angle 63.363\ A$

phase angle $\phi = 63.363°$

Power factor $= \cos\phi$

$= \cos 63.363$

$= 0.448$ (leading)

$V = IR$

$V = IZ$

Now,

$Z_{coil} = R + jX_L$

$Z_{coil} = (32 + j15.7)\ \Omega$

Imp $V_{coil} = I \times Z_{coil}$

$V_{coil} = (7.005 \angle -63.363)(32 + j15.7)$

$V_{coil} = (249.685 \angle -37.229)\ V$

Power, $P = |V|\,|I| \cos\phi$

$P = 500 \times 7.005 \times 0.448$

$P = 1569.12\ W$

$P = I^2 R$

$= 7.005^2 \times 32$

$= 1569.12$

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The voltage applied to the circuit is $V=120\sin 314t$ and current is $I=100\sin(314t+\frac{\pi}{6})$. Calculate the value of resistance and capacitance which are connected in the circuit. Also determine the power consumed by the circuit. Draw phasor diagram

⇒ Given:

R-C ckt.

$V = 120\sin 314t$

$I = 100\sin(314t + \frac{\pi}{6})$

$R = ?$

$C = ?$

$P = ?$

Phasor diag. = ?

⇒ We have,

$V = 120\sin 314t$

compare it with

$V = V_m \sin \omega t$

$\therefore$ $V_m = 120\,V$, $\omega = 314$

Also, we have,

$I = 100\sin(314t + \frac{\pi}{6})$

compare it with

$I = I_m \sin(\omega t \pm \phi)$

$\therefore$ $I_m = 100\,A$, $\phi = \frac{\pi}{6} = 30^\circ$

Now, we know that,

$I = \frac{V}{Z}$, $Z = \frac{V}{I} = \frac{120}{100}$

$Z = 1.2\,\Omega$

Now,

$\cos\phi = \frac{R}{Z}$

$\therefore R = Z\cos\phi = 1.2\cos 30^\circ$

$R = 1.039\,\Omega$

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As ϕ is +ve,
∴ Given ckt is R-C ckt.

$$\therefore Z = \sqrt{R^2 + X_C^2}$$

$$\therefore X_C = \sqrt{Z^2 - R^2}$$

$$= \sqrt{1.2^2 - 1.039^2}$$

$$X_C = 0.6\ \Omega$$

$$\frac{1}{\omega C} = 0.6$$

$$\frac{1}{314 \times C} = 0.6$$

$$\therefore C = \frac{1}{0.6 \times 314}$$

$$C = 5.307 \times 10^{-3}\ F$$

$$C = 5307\ \mu F$$

Now, Power, $P = |V||I| \cos\phi$

$$= V_{rms} \times I_{rms} \times \cos\phi$$

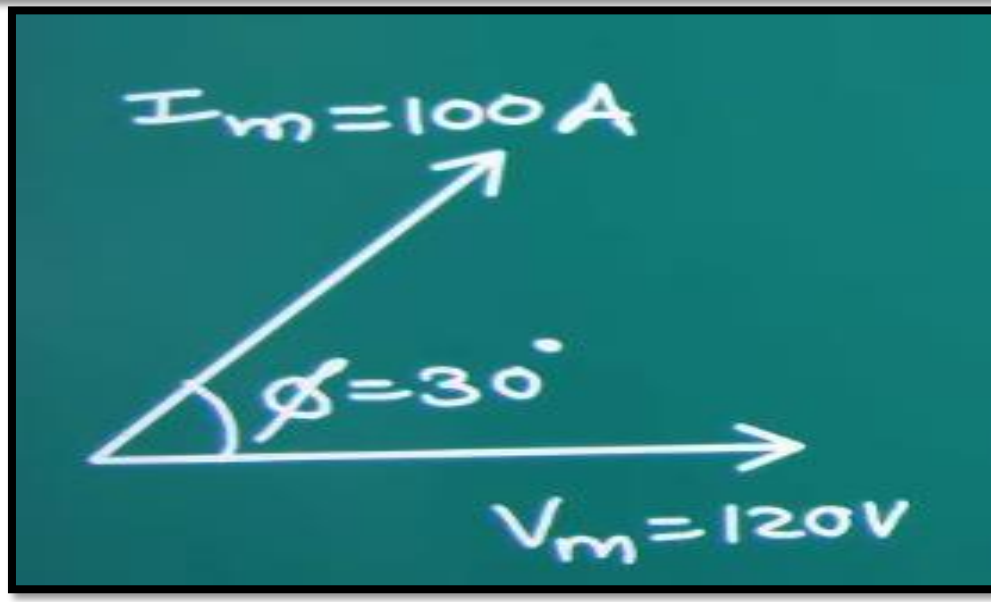
$$= \left(\frac{V_m}{\sqrt{2}}\right)\left(\frac{I_m}{\sqrt{2}}\right) \times \cos\phi$$

$$= \left(\frac{120}{\sqrt{2}}\right)\left(\frac{100}{\sqrt{2}}\right) \cos 30^\circ$$

$$P = 5196.152\ W$$

$$P = 5.196\ kW$$

Phasor diagram



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A coil having R=20Ω, L=0.2H connected in series with 50 uF capacitor across 250V, 60Hz A.C. supply.

Determine :

1) Current
2) Power factor
3) Active power
4) Voltage across capacitor.

Given :

R-L-C ckt

$R = 20\,\Omega$

$L = 0.2\,H$

$C = 50\,\mu F = 50 \times 10^{-6}\,F$

$V = 250\,V$

$f = 60\,Hz$

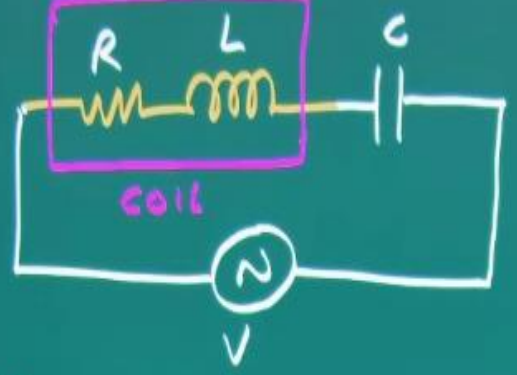
To Find :

i) $I = ?$

ii) $\cos\phi = ?$

iii) $P = ?$

iv) $V_C = ?$



$X_L = 2\pi f L = 2\pi \times 60 \times 0.2$

$X_L = 75.398\,\Omega$

$X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi \times 60 \times 50 \times 10^{-6}}$

$X_C = 53.051\,\Omega$

For R-L-C ckt,

$Z = R + j(X_L - X_C)$

$= 20 + j(75.398 - 53.051)$

$Z = 20 + j22.347\ \Omega$

$Z = 29.989 \angle 48.172\,\Omega$

Now, $I = \frac{V}{Z} = \frac{250}{29.989 \angle 48.172}$

$\therefore I = 8.336 \angle -48.172\,A$

Power factor, $P.f. = \cos\phi$

$\therefore P.f. = \cos 48.172$

$P.f. = 0.666$ (lagging)

Active Power, $P = |V|\,|I| \cos\phi$

$P = 250 \times 8.336 \cos 48.172$

$P = 1389.812\,W$

$P = 1.389\,kW$

Voltage across capacitor V_C,

$V_C = I \times X_C$

$= 8.336 \times 53.051$

$V_C = 442.233\,V$

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A coil of inductance 0.0805 H takes a current of 5 A when connected in series with 50 μF across 240V, 50Hz A.C. supply. Determine :

1) Resistance of coil
2) Power factor
3) Active power
4) Voltage across coil
5) Power factor of coil

Given :

$L = 0.0805\ H$

$I = 5\ A$

$C = 50\ \mu F = 50 \times 10^{-6}\ F$

$V = 240\ V$

$f = 50\ Hz$

To Find :

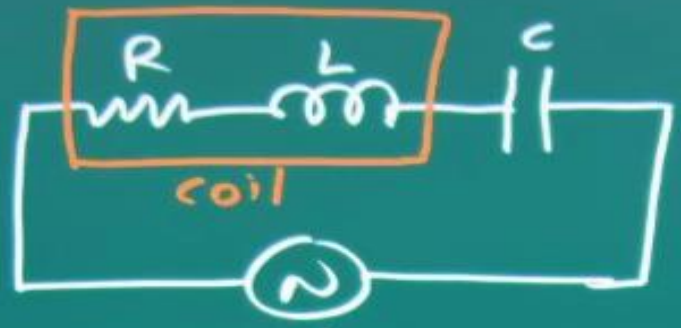
i) $R_{coil} = ?$

ii) $\cos\phi = ?$

iii) $P = ?$

iv) $V_{coil} = ?$

v) $\cos\phi_{coil} = ?$



$X_L = 2\pi f L$

$= 2\pi \times 50 \times 0.0805$

$X_L = 25.289\ \Omega$

$X_C = \frac{1}{2\pi f C}$

$= \frac{1}{2\pi \times 50 \times 50 \times 10^{-6}}$

$X_C = 63.661\ \Omega$

$I = \frac{V}{Z}$

$\therefore Z = \frac{V}{I} = \frac{240}{5} = 48\ \Omega$

$\therefore Z = 48\ \Omega$

For RLC ckt,

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$\therefore R = \sqrt{Z^2 - (X_L - X_C)^2}$$

$$R = \sqrt{48^2 - (25.289 - 63.661)^2}$$

$$\therefore \boxed{R = 28.837\ \Omega} \checkmark$$

Power factor, $\cos\phi = \frac{R}{Z}$

$$= \frac{28.837}{48}$$

$$\therefore \boxed{\cos\phi = 0.6} \text{ (leading)}$$

Power, $P = |V|\ |I| \cos\phi$

$$= 240 \times 5 \times 0.6$$

$$\boxed{P = 720\ W} \checkmark$$

$$Z_{coil} = \sqrt{R^2 + X_L^2}$$

$$= \sqrt{28.837^2 + 25.289^2}$$

$$\boxed{Z_{coil} = 38.355\ \Omega}$$

$$V_{coil} = I \times Z_{coil}$$

$$= 5 \times 38.355$$

$$\therefore \boxed{V_{coil} = 191.775\ V}$$

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